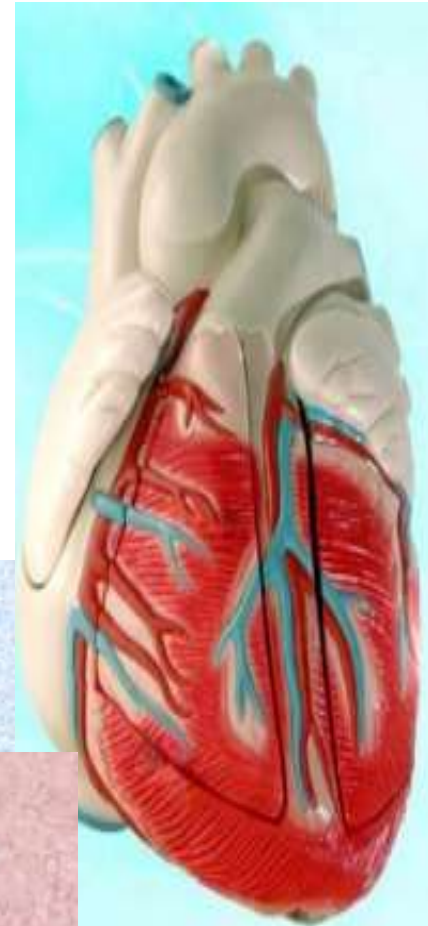




Biochemistry of cvs MODULE

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**Aug. 2024GC
JIMMA ETHIOPIA**

Outline

- Cardiac energetics
- Characteristics of fuel metabolism in the heart
- Biomarkers of cardiac injury and determination of cardiac biomarkers
 - ✓ **AST, CK, LDH,**
 - ✓ **Troponin ,Mb and NT-proBNP**
- Lipid metabolism
- Hyperlipidemia & Atherosclerosis
- Familial hypercholesterolemia (FH)
- Hypertriglyceridemia

Cardiac energetics

- Cardiac muscle is highly efficient in generating ATP, for continuous pumping action
- For normal cardiac function, there is tight coupling of ATP production and myocardial contraction
- Cardiac muscle has metabolic flexibility based on energy demands
- **Fatty acids** are the preferred source of energy for cardiac muscle under normal conditions

Characteristics of fuel metabolism in the heart

- Heart use all classes of energy substrates, including lipids, carbohydrates, amino acids, and ketone bodies, for ATP production in the mitochondrion
- Mitochondria occupy one third of the volume in cardiac myocytes,
- The rebusting of cardiac metabolism is reflected by,
 - its highest oxygen consumption rate per unit weight basis

The Major Sources Energy to the heart is Fatty acid

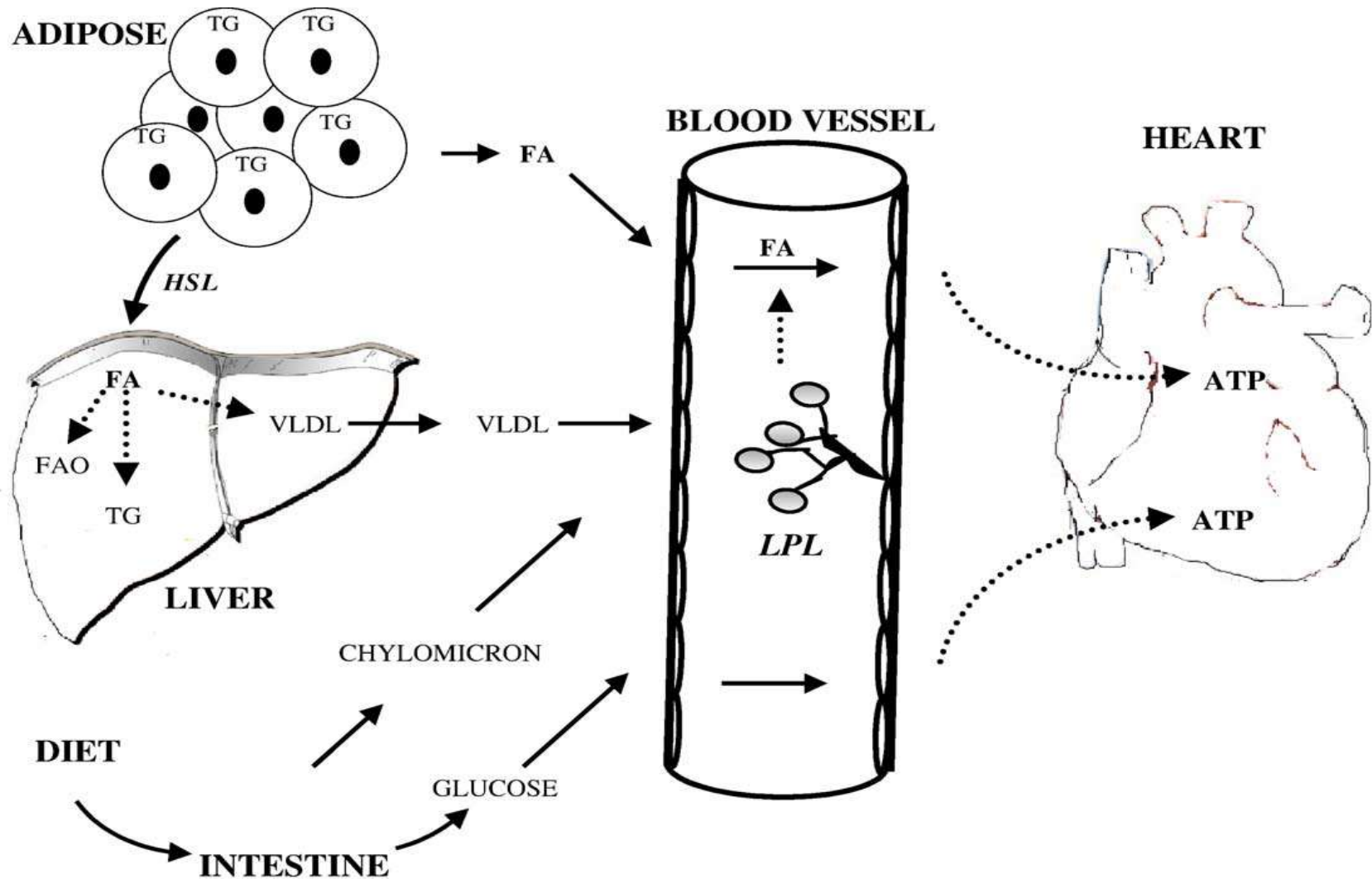


Fig: Source of energy for heart /Cardiac

Cardiac Muscle Metabolism

- Optimal cardiac function depends on
 - ✓ the **efficient matching of energy generating to energy expenditure**
- Heart described as a **metabolic omnivore**, b/c
 - it has the capacity to oxidize fat and carbohydrates for its energy needs
- Cardiac muscles dependent upon
 - **adequate availability oxygenation and d/f substrates such as:**
 - ❖ Fatty acids (60-70%) & carbohydrates (25%)
 - ❖ 3- 5% from amino acids and ketone bodies.

Cardiac Muscle Metabolism Cont...

- The metabolic machinery of the heart utilizes oxygen up to
- 80%-90% of the maximum capacity of oxphos; however, at rest, the heart operates at 15-25% of its maximal oxidative capacity,
 - b/c under resting conditions activity of a key metabolic enzyme can greatly reduce flux through the pathway
- This is because flux through metabolic pathways can be rapidly turned on or off,
- Depend on concentration of inhibitory or stimulatory metabolites through the pathway w/c regulated
 - via modification of regulatory enzymes
- These mechanisms allow for rapid adaptation to acute stresses such as:
 - exercise, ischemia and fasting

Cardiac Muscle Metabolism Cont...

- The cardiomyocytes is the most energy hungry muscle in the body and needs a constant supply of ATP
- it show an elevated rate of ATP hydrolysis, which is strongly coupled to oxidative phosphorylation because under aerobic conditions,
 - over 95% of these cells' ATP is produced in this process
- This is why mitochondrial integrity is so important for heart function

Cardiac Muscle Metabolism Cont...

- Each day, the heart beats about 100,000 times
 - ✓ it pumps 5L of blood per minute, 300 L per hour
- It is estimated that **over 6 kg of ATP** is hydrolysed by the heart daily for the pumping function

The primary energy source for cardiac muscles is ATP

- In Cardiomyocytes uses ATP's :
 - 60-70% for **muscle contraction**
 - 30 - 40% for the **sarcoplasmic reticulum (SR) Ca^{2+} -ATPase and other ion pumps**
- A decrease in cardiac metabolism is linked to a higher risk of heart failure

Cardiac Muscle Metabolism Cont...

- In order to maintain efficiency heart activities:
 - the enzymes, and
 - structural elements of the heart membranes undergo constant turnover
- Every 30 days, the entire heart is
 - ✓ reconstructed with new protein components
- Energy yield per gram of substrate metabolized, is more with fatty acids,
 - 37kJ/g versus
 - 16kJ/g with carbohydrates However,
- this is at the cost of much greater amount of oxygen consumed,
 - i.e. 2.0L/g with fats versus 0.83L/g with carbohydrates

Cardiac Muscle Metabolism Cont...

- Heart needs more energy than any other organ in the body
- Heart contraction consume **about 15%** DEE of the body
- It depend **on aerobic metabolism** for their energy needs
 - **b/c they contain many mitochondria in cardiomyocyte space and rich in myoglobin**
- However, storage of energy in the heart is very limited with regard to the organ's needs (very little glycogen)
- Glycogen pool in the heart is relatively small (~30 mmol/g wt compared with ~150 mmol/g wt in skeletal muscles).
- reserved energy in cardiomyocyte , **glycogen ensure only for about 20 seconds of normal activity**

Mec. Glucose Transported into the cells

- Heart muscle, skeletal muscle & Adipose tissue:
 - GLUT-4 have w/c dependent on Insulin
- exercise can independently increase GLUT4 translocation to the cell membrane without the aid of insulin
- The mechanism involves:
 - recruitment of GLUT-4 from IC vesicles into the plasma membrane
- insulin independent organs
 - brain , RBCS, intestinal cells & kidneys

Transport of Glucose to Heart Muscles & Use of Insulin

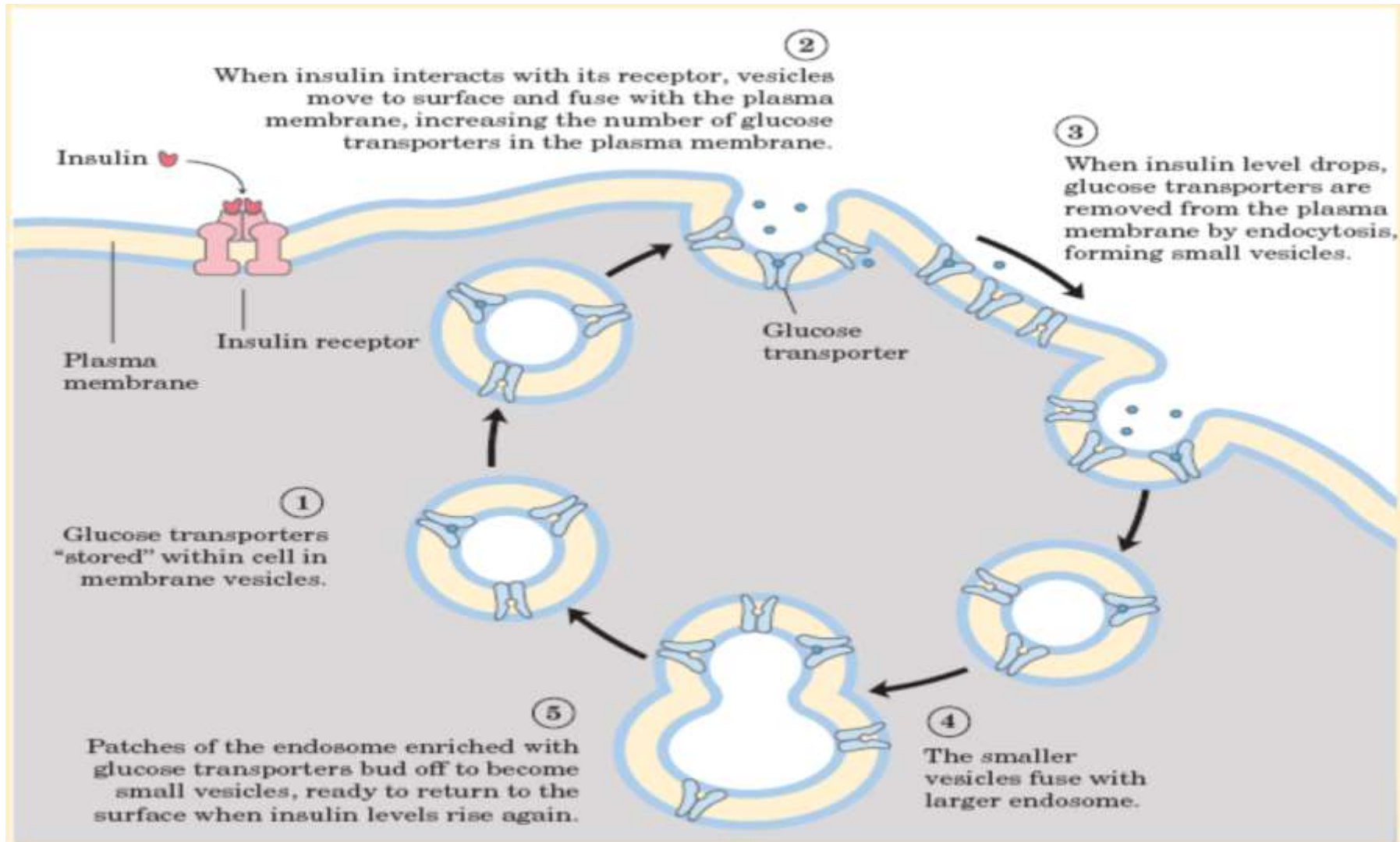


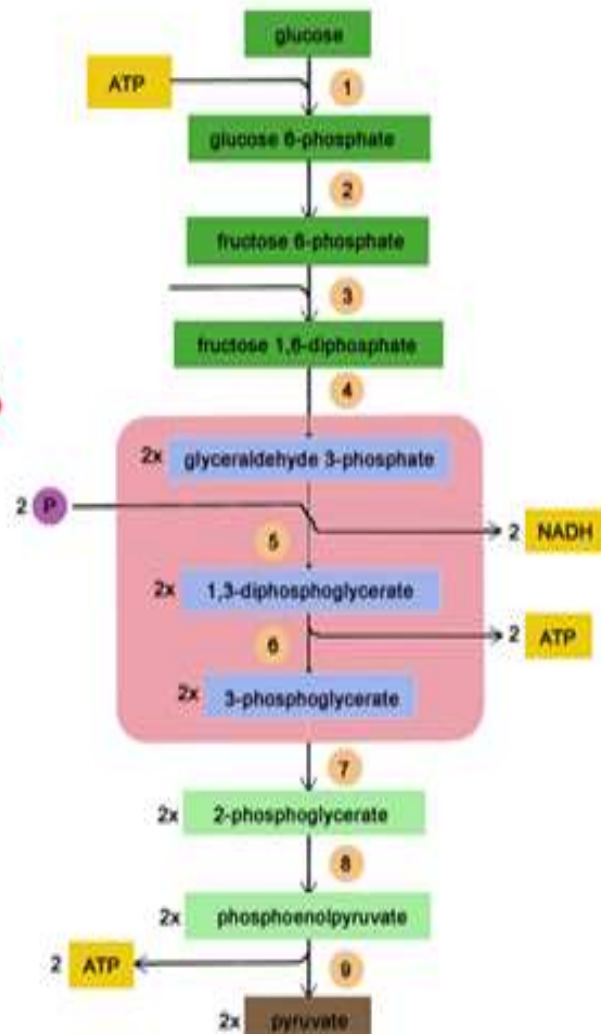
FIG-Special Regulation by insulin of glucose transport by GLUT- 4 into a myocyte

Transport of Glucose to the Heart Muscles

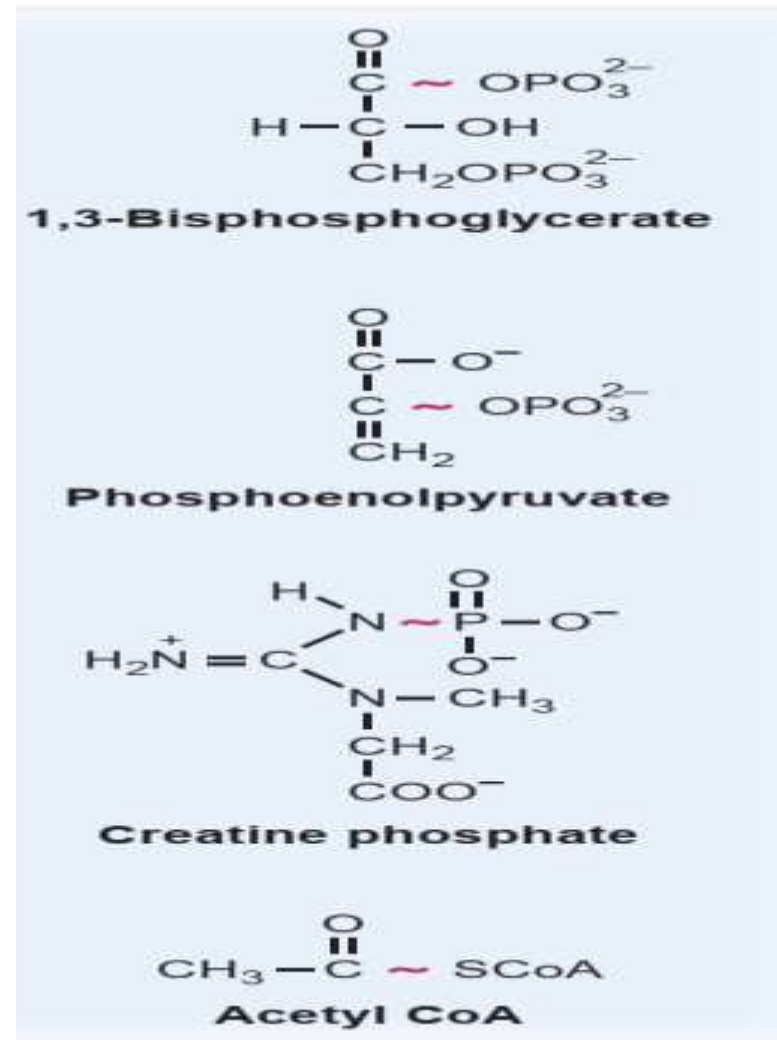
- **GLUT1** which is insulin independent and predominate in
 - **hypertrophied, atrophied and failing myocardium**
- GLUT1 is the first gene whose transcription is **stimulated**
 - **in response to hypoxia and inhibit oxidative phosphorylation**
- There are fundamentally two mechanisms for ATP synthesis:
 - Substrate level phosphorylation
 - Oxidative phosphorylation (accounts for > 95% of the ATP synthesized in the heart mitochondria)

Substrate level phosphorylation (from high energy bonds)

Anaerobic & Aerobic
Glycolysis

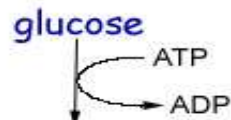


1. glucokinase, hexokinase. 2-gluc-6-p isomerase. 3-PFK 4-aldolase. 5-glyceraldehyde 3 p dehydrogenase. 6-phosphoglycerate kinase. 7-phosphoglycerate mutase. 8-enolase. 9-pyruvate kinase.



Anaerobic & Aerobic Glycolysis

Hexokinase
Glucokinase



glucose 6-phosphate

fructose 6-phosphate

isomerase

Phosphofructokinase 1
(PFK 1)



Fructose 2,6 bisphosphate

Phosphofructokinase 2 (PFK 2)
w/c activate PFK1

glyceraldehyde 3-phosphate

dihydroxyacetone phosphate

Aldolase

Pi + NAD

NADH+H

1,3-bisphosphoglycerate

Glyceraldehyde 3 Phosphate
Dehydrogenase

ADP

3-phosphoglycerate

Phosphoglycerate Kinase

2-phosphoglycerate

Mutase

phosphoenolpyruvate

ADP

ATP

pyruvate

Enolase

NADH + H

NAD

Lactate Dehydrogenase

lactate

Pyruvate Kinase

GAPDH is major regulatory step,
it is inhibited by accumulation of
NADH and activated by NAD+.

Pyruvate

Pyruvate Dehydrogenase

Acetyl CoA

TCA or Fatty Acid Synthesis

Severe ischemia in heart → lactate and NADH
accumulation → cessation of oxidative metabolism
and lactate production.

production

Hormonal regulation of cardiac metabolism

- Glycogen concentrations are increased by an elevated supply of exogenous substrate w/c is followed with hyperinsulinemia
- **Insulin** promotes glucose uptake and utilization by cardiac muscle cells
- increases Glc availability in adipose tissue used for the synthesis of Fatty acids &
- Increase Glc, in the glycolytic pathway provide glycerol to adipose for TAG synthesis
- **Glucagon** stimulates fatty acid release from adipose tissue and fatty acid oxidation in the heart

Energy Metabolism in Normal Aerobic Heart

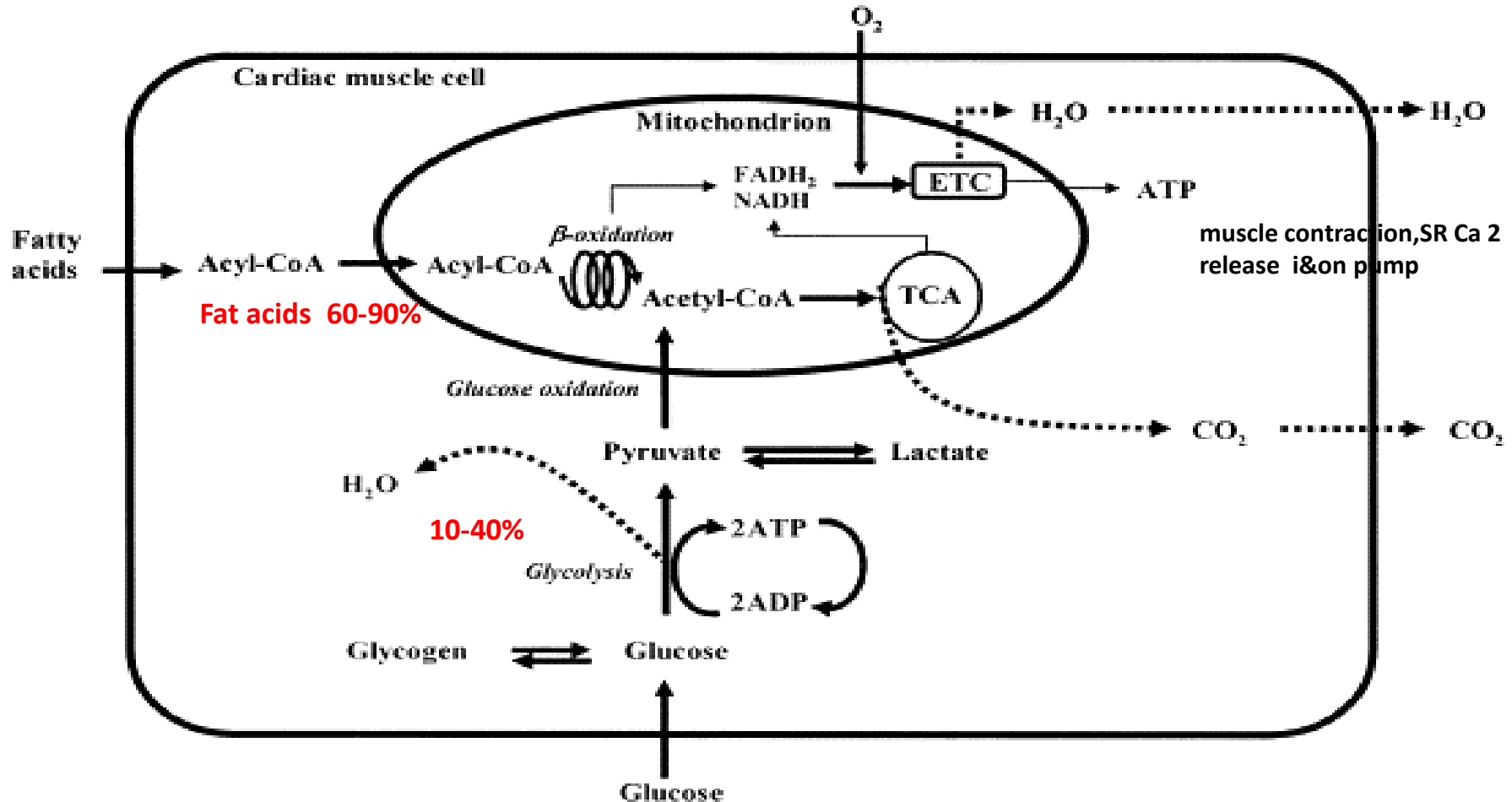
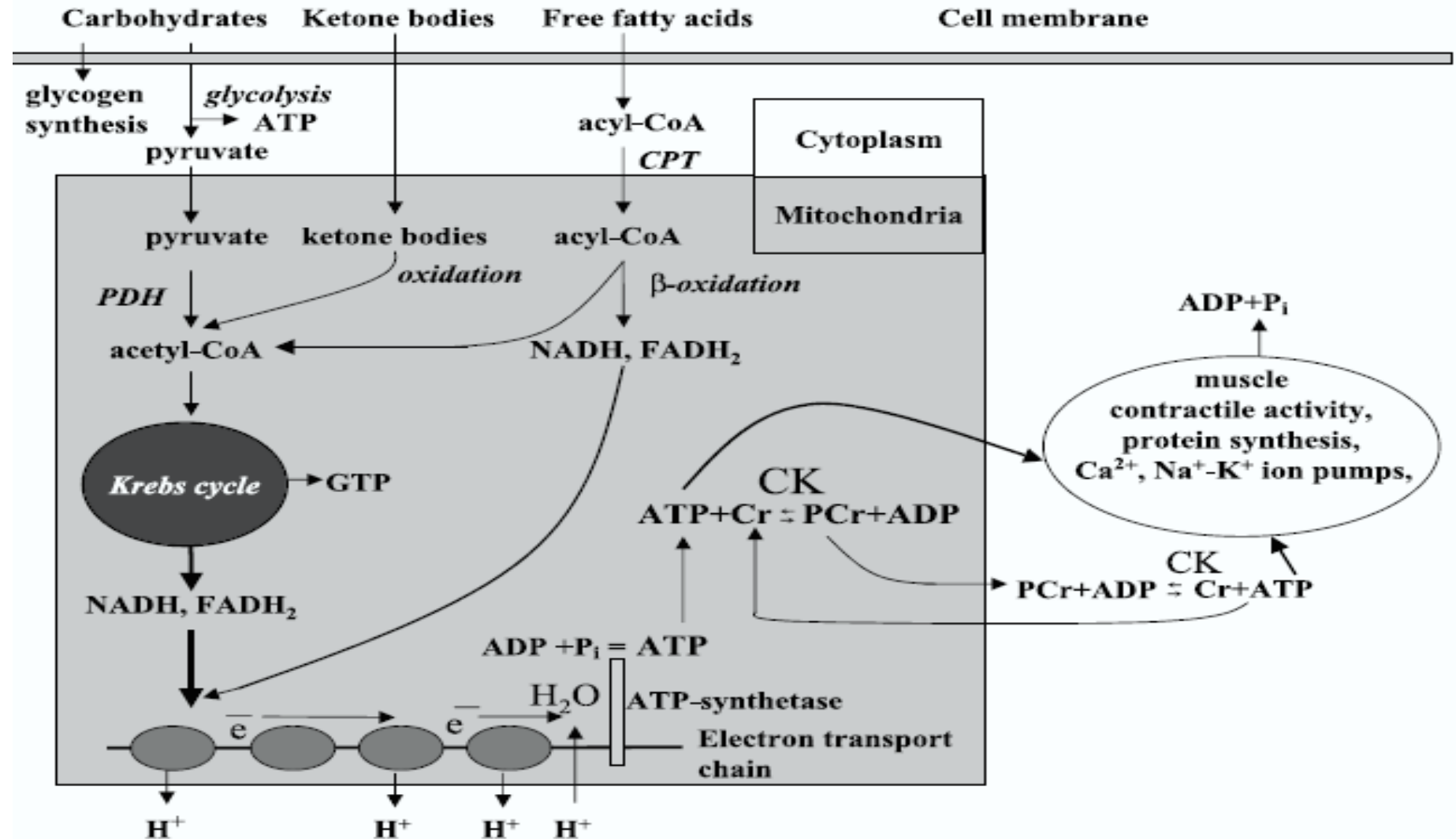


Fig : TCA is fueled by acetyl-CoA formed by oxidative decarboxylation of pyruvate (glycolysis) (10-40%) + from β -oxidation of FA (60-90%).

The reducing equivalents: NADH and FADH₂ (generated by glycolysis, oxidation of lactate, pyruvate and β -oxidation of FAs) deliver electrons to ETC $\rightarrow$ ATP (oxidative phosph).

ATP Production and its Utiliserions in Cardiac Muscles Mainly through Oxidative Phosphorylation (95%)

Figure. Normal Myocardial Metabolism



Cardiac Muscle Metabolism Cont...

- ❖ Mitochondrial oxidative phosphorylation is fueled with energy from electrons, that generate from NADH and FADH₂ produced primarily in the
 - ✓ Glycolysis, TCA cycle & β -oxidation of Fatty acid
- ✓ In the healthy heart ,the rate of oxidative phosphorylation process (ATP synthesis) is competently linked to rate of ATP hydrolysis ,
so that ATP content remains constant ,even with large increases in cardiac power, such as occur during intense exercise or stress

Cont...

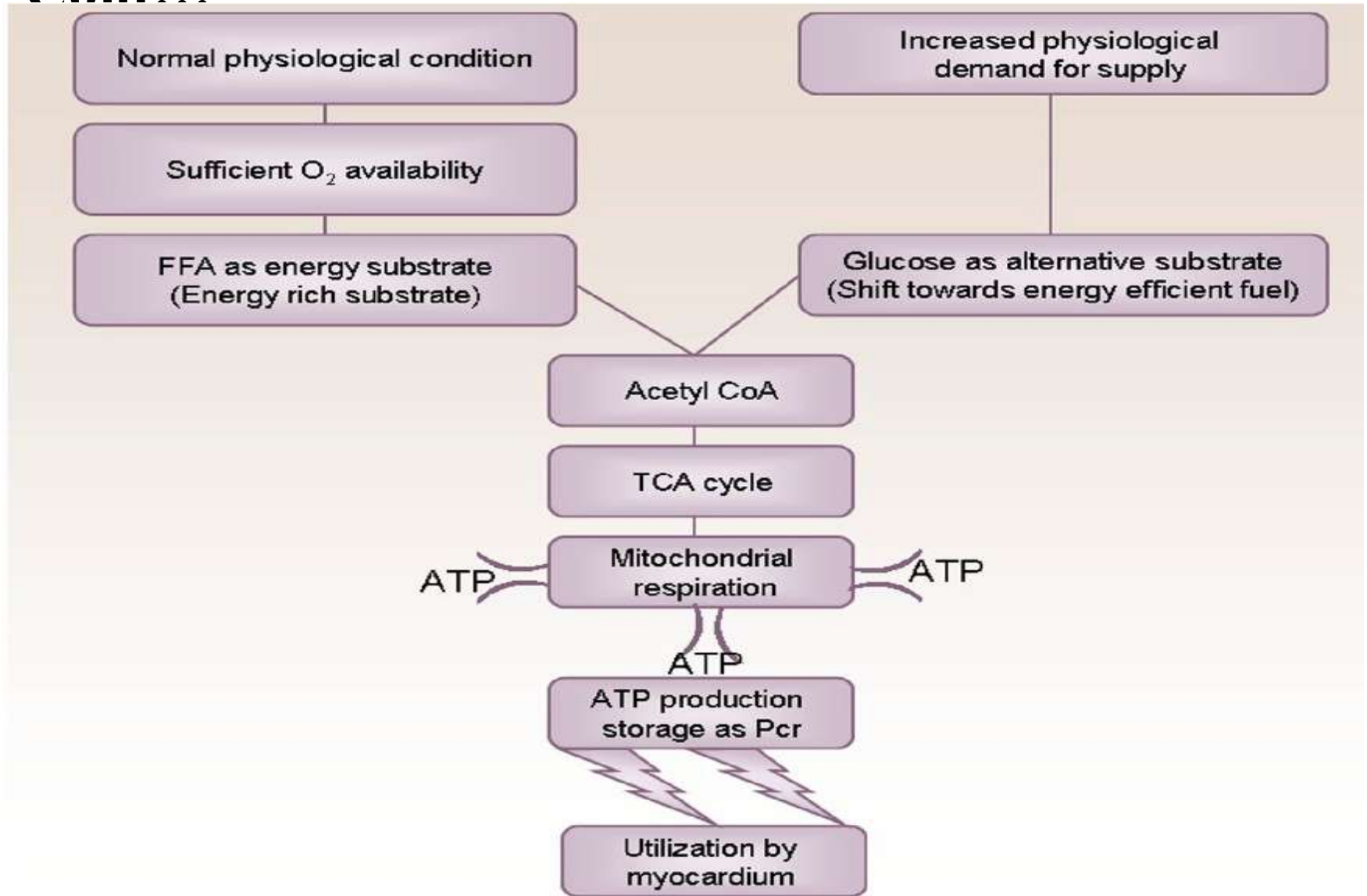


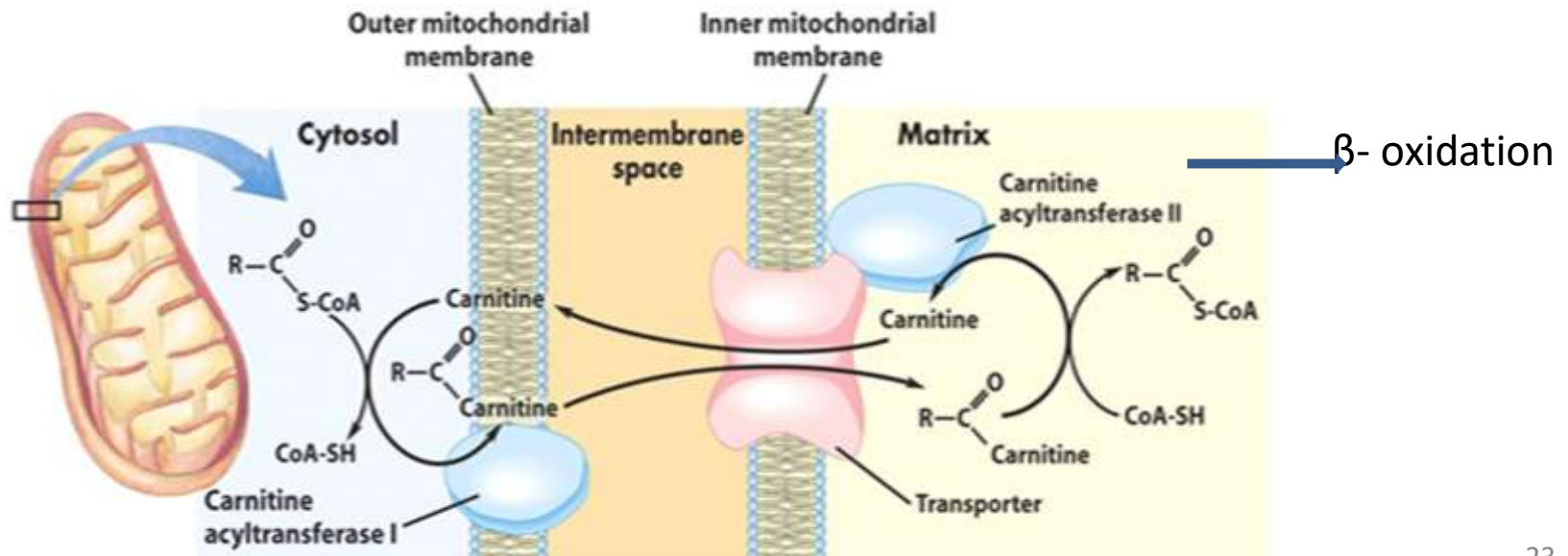
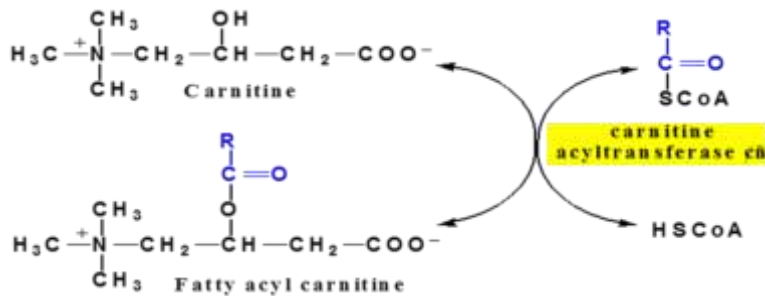
Fig. 2. Myocardial metabolism in normal heart. .. Pcr, Phosphocreatinine

Energy Metabolism in Normal Aerobic Heart

- Under normal conditions, the Cardiac muscle predominantly relies on **aerobic metabolism to generate ATP**
 - **Via converts the chemical energy stored in fatty acids and glucose into ATP**
- The oxidation of glucose to pyruvate and the activity of PDH in the heart are decreased by
 - elevated rates of fatty acid oxidation
- The oxidation of pyruvate is regulated at the pyruvate dehydrogenase (PDH) reaction.

β -oxidation fatty acids

- FAs enter the cardiomyocyte by: Passive diffusion
- β -oxidation fatty acids: carnitine transport long-chain fatty acids (LCFA) into mito.



β-oxidation of fatty acids

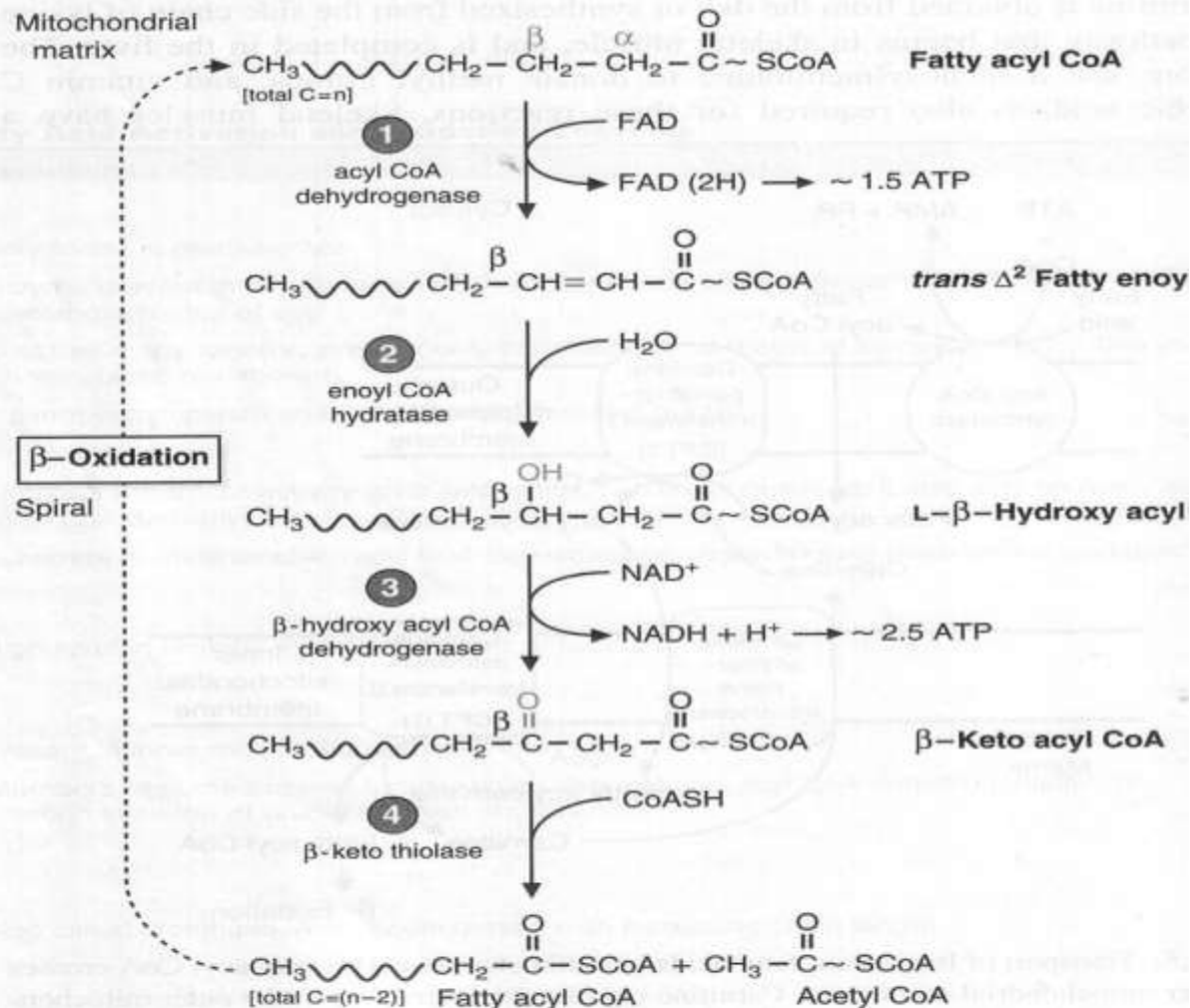
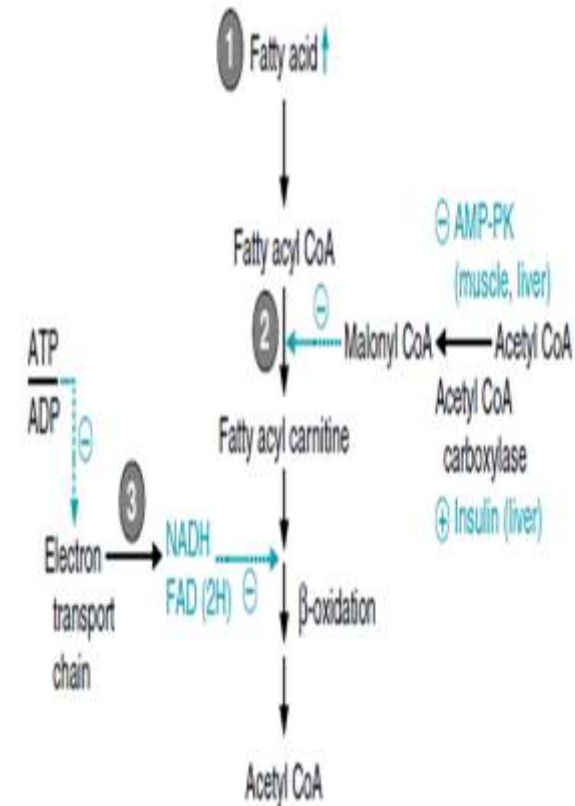


Fig. 23.7. Steps of β-oxidation . The four steps are repeated until an even-chain fatty completely converted to acetyl CoA. The FAD(2H) and NADH are reoxidized by the e transport chain, producing ATP.



Entry of long chain acyl-CoA into mitochondrion is a regulated with the rate-limiting step at the muscle form of CPT1 reaction CPT-I can be strongly inhibited by malonyl CoA (on cytosolic side of the enzyme).

- Malonyl -CoA - key physiological regulator of FA oxidation in heart (decrease in malonyl -CoA $\rightarrow$ inc. FAs uptake and oxidation).
- Formed from the carboxylation of acetyl-CoA (acetyl- CoA carboxylase – ACC) from extramitochondrial acetyl- CoA (derived from citrate via ATP- citrate lyase reaction)

Metabolic network

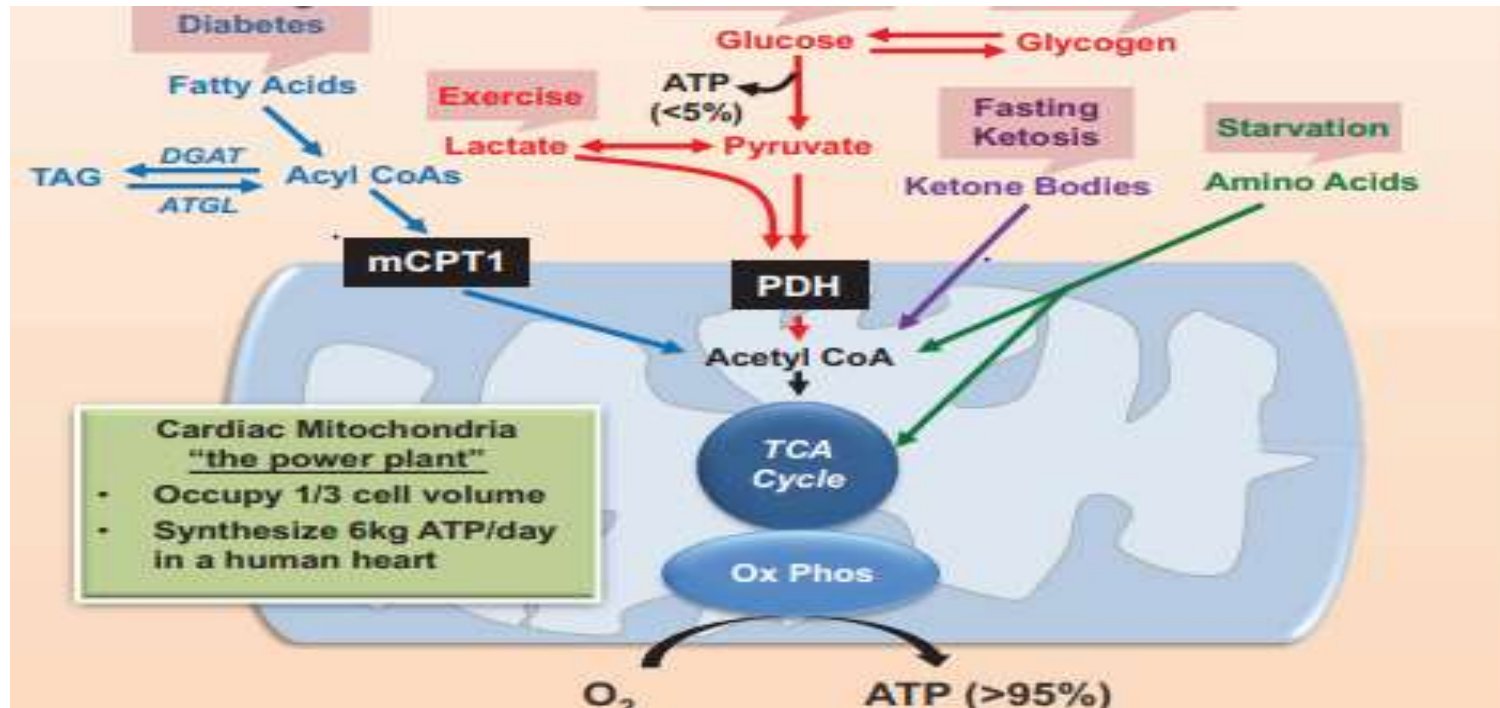


Fig 1. The metabolic network .

- The energy-yielding substrates (fatty acids, glucose, ketones, and amino acids), via specific catabolic pathways, converge on acetyl-CoA production with subsequent entry into the TCA cycle.
- The final step of energy transfer is accomplished through oxidative phosphorylation (OxPhos), supplying >95% of ATP consumed by the heart.
- ATGL indicates adipose triglyceride lipase; DGAT, diacylglycerol acyltransferase

Metabolic network ...

- Fatty acids are the predominant substrate used in the adult myocardium. However, the cardiac metabolic network is highly flexible in using other substrates when they become abundantly available .

Starvation

- **Prolonged fasting** or ketogenic diet increases the blood level of ketone bodies and results in enhanced use by the heart
 - these significantly reduces the oxidation of glucose and fatty acids

Metabolic network ...

Sources energy during exercise

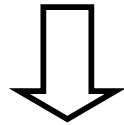
- Cardiac energy is depleted quickly, especially when exercising
- However, lactate works to pick up the slack during cardiovascular exercise
 - the heart uses lactate and glucose as more oxygen efficient substrates
- The ability to switch between energy sources (fatty acids, glucose, lactate, ketones) is called metabolic substrate flexibility and
 - **is essential to meet the fluctuating demands on the heart during exercise**

Cardiac Muscles Produce ATP Mainly through Oxidative Phosphorylation (95%)

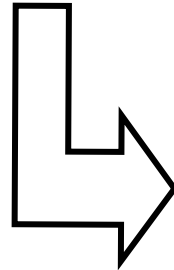
Glucose oxidation

- Provides 10% - 40% of energy

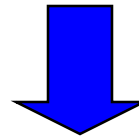
Glucose



Pyruvate



Acetyl coA

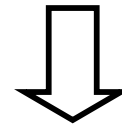


Energy (ATP)

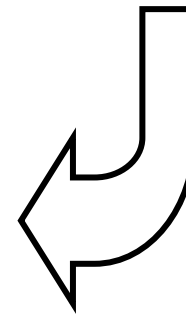
Fatty Acid β -oxidation

- Provides 60% to 90% of energy

Fatty acids



Acyl coA



For a human heart, the amount of ATP turned over rate during a 1-day period is 15 to 20 times of its own weight

Oxidative Phosphorylation by normal heart

Metabolic network ...

- ❑ Some metabolic changes that come from an increase in cardiovascular endurance

Include :

- A larger quantity and size of mitochondria cells
- An increase in skeletal muscle myoglobin concentration
- A greater ability to oxidize fatty acids for energy at rest.
- An increase in stores of glycogen
- Regular cardiac exercise is another excellent way to improve your metabolism's functions and cardiovascular health
- because
 - it helps your heart have more energy to consume.

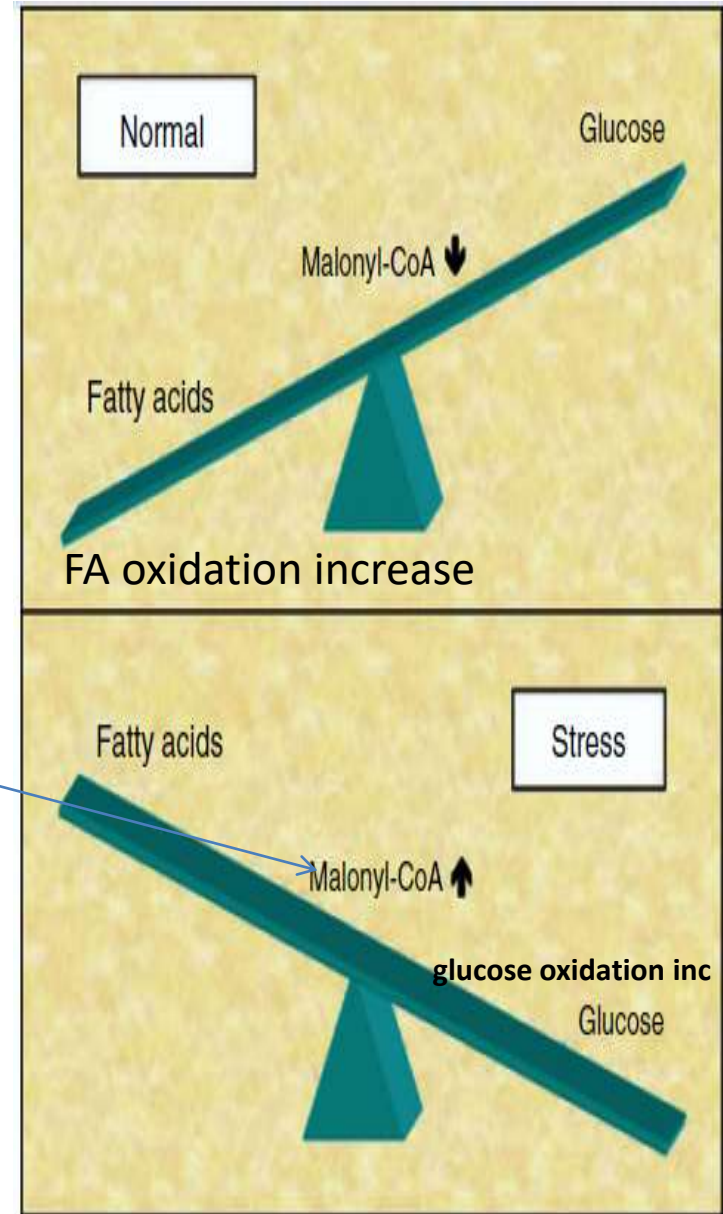
Pathological Hypertrophy and Failure

- Metabolic remodeling in hypertrophied heart is associated with decreases in the overall ATP synthesis by oxidative metabolism
- hypertrophy, is characterized by
 - increased reliance on glucose metabolism and decreased FAO
- Although glycolysis is increased, its contribution to total ATP synthesis is limited as glycolytic ATP accounts for < 5% of total energy used by the heart
- Furthermore, glucose entry in the adult heart muscle is controlled by insulin; the coexisting insulin resistance in heart failure would limit the glucose availability and hence compromise the capacity for ATP synthesis

Biochemistry of Heart Failure (HF)

Heart Failure (HF)

- ❑ Heart failure reduces the capacity to transfer the energy from food stuff into ATP
- ❑ In the advanced stage of HF
 - down regulation in FA oxidation , increase malonyl-coA level
 - Increased glucose oxidation via glycolysis & reduced respiratory chain activity



Pathological Hypertrophy ...

- Although only 5 % the heart's ATP is produced in glycolysis, but it becomes very important under anaerobic or ischemic status
- Indeed, **in heart failure and hypertrophy**, there is a metabolic switch towards favoring carbohydrate over FAs metabolism in the heart
- insulin-independent glucose transporter (GLUT1) showed substantial increases of glucose uptake and glycolysis but maintained normal cardiac function and lifespan, suggesting that increased glucose use does not harm the adult heart in the long term.

Pathological Hypertrophy ...

- It is known that pathological cardiac hypertrophy is associated with depletion of energy reserves
- manifested as maintained ATP levels but a reduction of the energy reserve compound, phosphocreatine (PCr).

Pathological Hypertrophy ...

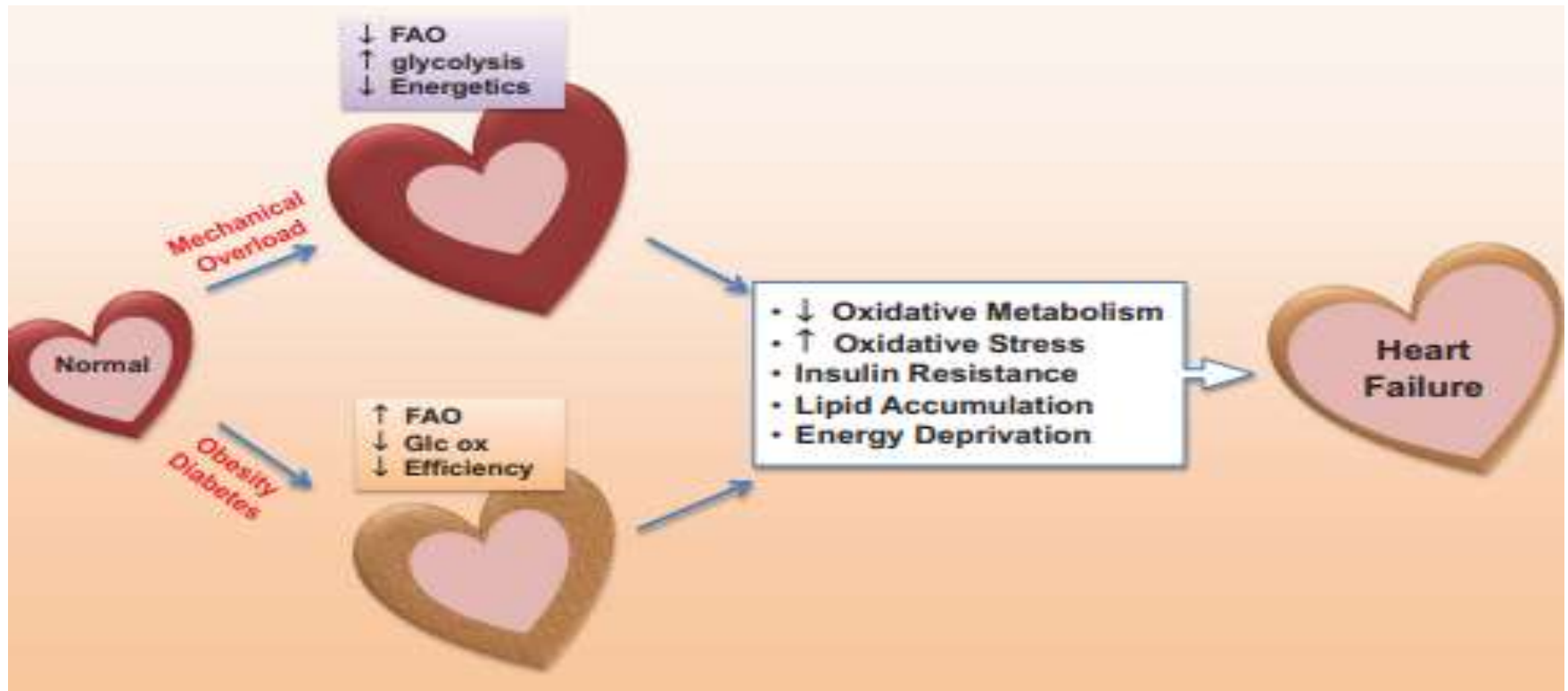


Figure 3. Metabolic remodeling and the development of heart failure.

for example, hypertension, valvular disease, or post-myocardial infarction, is accompanied by metabolic remodeling characterized by decreases in fatty acid oxidation (FAO) and increases in glycolysis

Pathological Hypertrophy ...

- A unique aspect of cardiac metabolism in obesity and diabetes mellitus is that the supply of substrates exceeds the need for ATP synthesis.
- Despite increased FAO, hearts of obese
- individuals and those with diabetes mellitus accumulate a significant amount of lipid
- . A positive correlation of cardiac lipid accumulation and cardiac dysfunction has been shown giving rise to the term lipotoxic cardiomyopathy

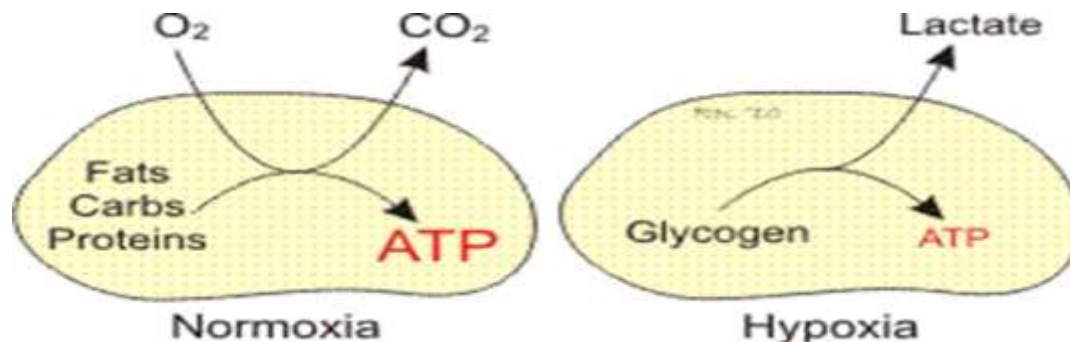
Ischemia and Heart Metabolism

In situations of ischemia:

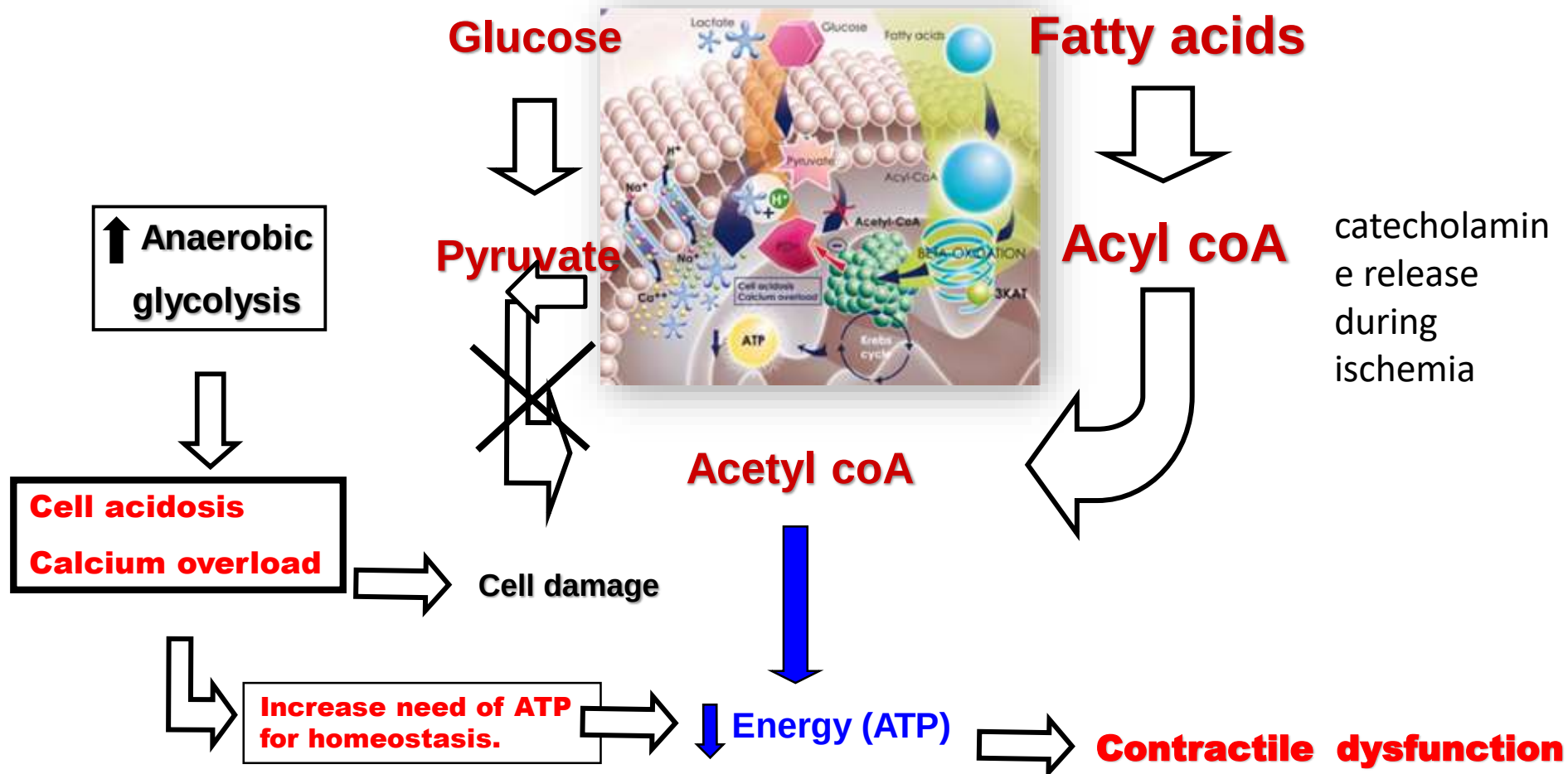
- Both oxidation of ,
 - fatty acid and glucose diminish due to oxygen shortage
- So anaerobic glycolysis becomes a more important source of energy as
 - it is the only process capable of producing ATP in the absence of oxygen However,
 - ATP generated by glycolysis is not sufficient to meet the energy needs of the beating heart

Ischemia and Heart Metabolism ...

- Moreover, in response to catecholamine release during ischemia , FFA levels increase and FFA oxidation becomes the preponderant residual in oxidative pathway
 - high levels of Acetyl-CoA produces which inhibit PDH and thus inhibit pyruvate oxidation
- The non oxidized pyruvate is converted into lactate & w/c release protons (H^+),this induce cellular acidosis and calcium overload , causing an increase in ATP expenditure in order to sustain cellular homeostasis



Cardiac Disease is Linked with Impaired Energy Metabolism



Therapeutic Metabolic Shifting In Failing Heart

- Shifting of cardiac metabolism from **FFAs** to **glucose** might be beneficial since;
 - ✓ **Fatty acid oxidation requires more oxygen;**
 - ✓ **Accumulation of products of fatty acid metabolism during ischemia contributes to the ischemic injury. Hence,**
 - **any approach that stimulates myocardial glucose oxidation and inhibits fatty acid oxidation** may be a solution

Therapeutic Metabolic Shifting In Failing Heart

- Metabolic interventions shifting the source of Cardiac energy toward carbohydrate utilization can different goals:
1. **Hypoxia lead to Inhibit free fatty acid oxidation** which **indirectly increases carbohydrate oxidation**
 2. **Increase glycolysis or carbohydrate utilization optimize energy metabolism in the heart & relieve ischemic symptoms, and improve cardiac efficiency/function.**

Cardiac Biomarkers and Cardiac Function Tests

Cardiac Biomarkers and Cardiac Function Tests

- ❖ The term Biomarkers (biological markers) means “any measurable and quantifiable biological parameters which contain:-
 - specific enzyme concentration,
 - hormone concentration
 - specific protein
- Biomarkers are a significant tool to diagnose disease conditions on time

Cardiac Biomarkers and Cardiac Function Tests

- Biomarkers is objectively measured :-
 - Normal biologic processes
 - Pathogenic processes or
 - Pharmacologic response to a therapeutic intervention
- Presence of such biological substances provide us information on health and pathophysiological status

Cardiac Biomarkers and Cardiac Function Tests

- Cardiac markers are biomarkers measured to evaluate heart function
- Cardiac biomarkers are substances that get released into the blood when the heart is damaged.
- Cardiac biomarkers can be detected in blood by a specialized immunoassay
- They are used to help diagnosis, evaluate, and monitor patients with suspected acute coronary syndrome (ACS)

IDEAL MARKER SHOULD HAVE DIAGNOSTICALLY

- i. High sensitivity (low detectable limit injury)
- ii. High specificity (absent in non-MI injury)
- iii. Rapid release at detectable concentration
- iv. Correlate efficiently with extent of MI injury
- v. Long $T_{1/2}$
- vi. Easy & inexpensive

TYPES OF CARDIAC BIOMARKERS

Enzymatic

- CK (CK-MB) – marker of injury
- LDH (LDH1) – marker of injury.
- AST– marker of injury.

Non enzymatic

- Troponins (T, I) – Injury marker
- Myoglobin – Injury marker.
- BNP (NT-proBNP) – marker of CHF.

TYPES CARDIAC MARKERS

1.CARDIAC ENZYME MARKERS

- i. CK
- ii. LDH
- iii. AST

2.CARDIAC PROTEIN MARKERS

- i. Myoglobin
- ii. Troponins
- iii. Pro-brain natriuretic peptide

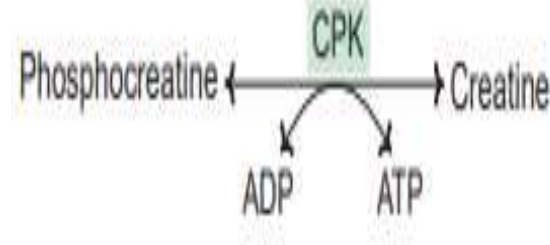
CARDIAC MARKERS

Enzymes in myocardial infarction

- The enzymes – namely creatine phosphokinase (CPK), aspartate transaminase (AST) and lactate dehydrogenase (LDH)
 - are important for the diagnosis of myocardial infarction (MI).
- 1. Creatine kinase (CK), formerly known as creatine phosphokinase, is an intracellular enzyme present in greatest amounts in
 - skeletal muscle, myocardium and brain
- Skeletal muscle expresses CK-MM (98%) & CK-MB (1%)
- myocardium expresses CK-MM at 70% and CK MB at ~30%

CARDIAC MARKERS...

- CPK catalyses the inter-conversion of creatine to phosphocreatine



- Creatine kinase (CK) has two subunits or polypeptide
 - composed of M (muscle) or B (brain), exists as three isoenzymes
- Each isoenzyme occurs as a dimer composed of two subunits M or B
- Isoenzymes : CK-MM, CK-BB and CK-MB both.

<i>Isoenzyme</i>	<i>Subunit</i>	<i>Tissue of origin</i>
CPK ₁	BB	Brain
CPK ₂	MB	Heart
CPK ₃	MM	Skeletal muscle

CARDIAC MARKERS...

- Creatine phosphokinase (precisely isoenzyme MB) is the first enzyme to be released into circulation within 6-18 h after the infarction
- Therefore, CPK estimation is highly useful for the early diagnosis of MI
- This enzyme reaches a peak value within 24-30 hs, and returns to normal level by the 2nd or 3rd day
- creatine kinase is not totally cardiac specific and exists also in skeletal muscles and other tissues
- The use of this test has decreased because of newer tests that have a better ability to detect heart damage only

CARDIAC MARKERS...

- ❑ Aspartate transaminase (AST or SGOT) rises sharply after CPK, and reaches a peak within 48 hours of the myocardial infarction.
- AST takes 4-5 days to return to normal level

Lactate dehydrogenase

- ❖ Catalysis the conversion of pyruvate to lactate.
- ❖ LDH-1 isozyme is normally found in the heart muscle
- ❑ Lactate dehydrogenase (LDH-1) generally rises from the second day after infarction, attains a peak by the 3rd or 4th day and takes about 10-15 days to reach normal level.
- ❑ Thus, LDH is the last enzyme to rise and also the last enzyme to return to normal level in MI

CARDIAC MARKERS...

2.CARDIAC PROTEIN

Troponins & Myoglobin

- Cardiac troponins (CT) : Although not enzymes, the proteins cardiac troponins are highly useful for the **early diagnosis of MI**
- the test measures the levels of troponin T or troponin I proteins in the blood
- These proteins are released when the heart muscle has been damaged or attack
- Cardiac troponin I (CTI) is released into circulation within four hours after the onset of MI,
 - reaches a peak value by 12–24 hours, and remains elevated for about a week

CARDIAC MARKERS...

- The more damage the heart muscle , the greater the amount of troponin T and I will be in the blood sample
- Cardiac troponin T (cTnT) and troponin I (cTnI) are cardiac regulatory proteins that control,
 - the calcium mediated interaction between actin and myosin
- Troponin I has been shown to be both a sensitive and specific marker of myocardial damage during cardiac surgery
- Cardiac troponin is the test of choice for finding a heart attack

CARDIAC MARKERS...

- Both cTnI and cTnT are highly specific for myocardial damage, but measurement of cTnI is generally preferred because
 - it becomes elevated earlier than cTnT, and it has been more widely reported in the veterinary literature
- Normal value are
- Troponin I < 0.3 ng/ml and Troponin T < 0.2 ng/ml.

CARDIAC MARKERS...

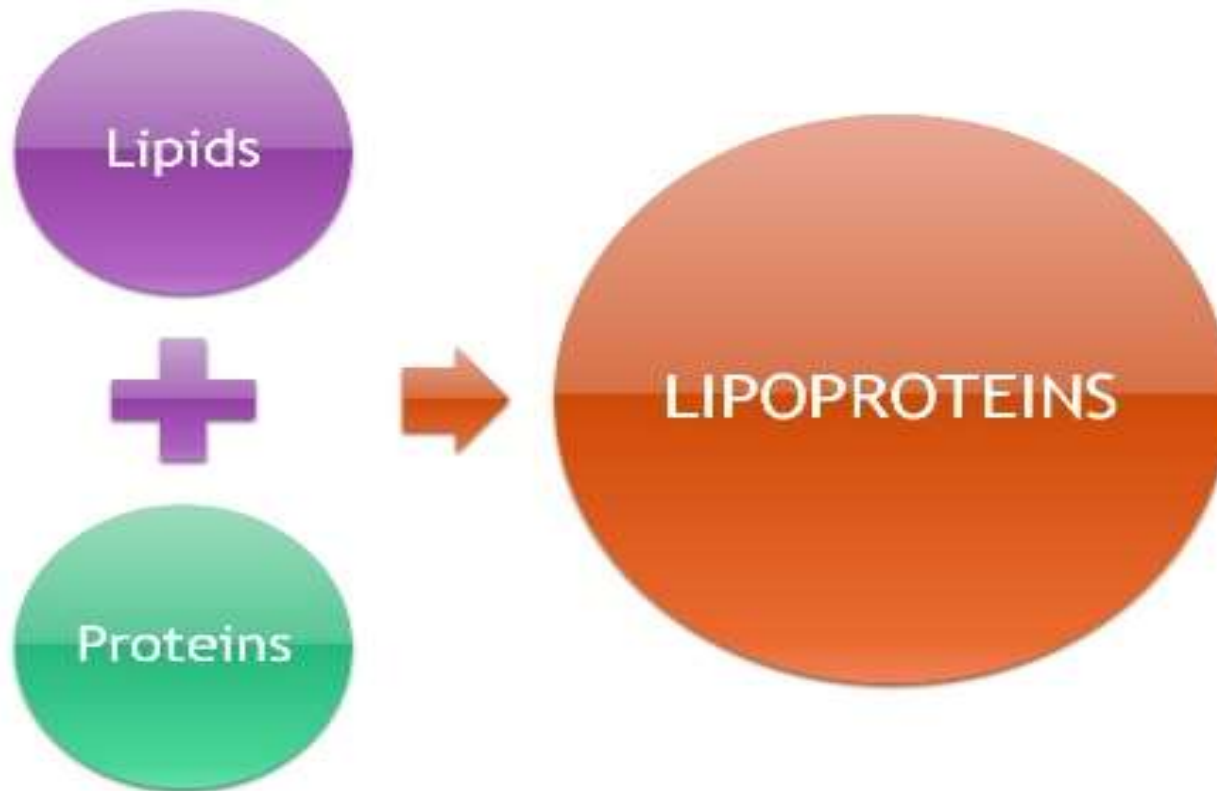
- Myoglobin is the primary oxygen-carrying pigment of muscle tissue
- The myoglobin is also an early marker for the diagnosis of MI. However,
 - **it is not specific to cardiac diseases**
- Myoglobin is used less than the other markers
- It is high when muscle tissue is damaged but it lacks specificity
- It has the advantage of responding very rapidly, rising and falling earlier than CK-MB or troponin.

CARDIAC MARKERS...

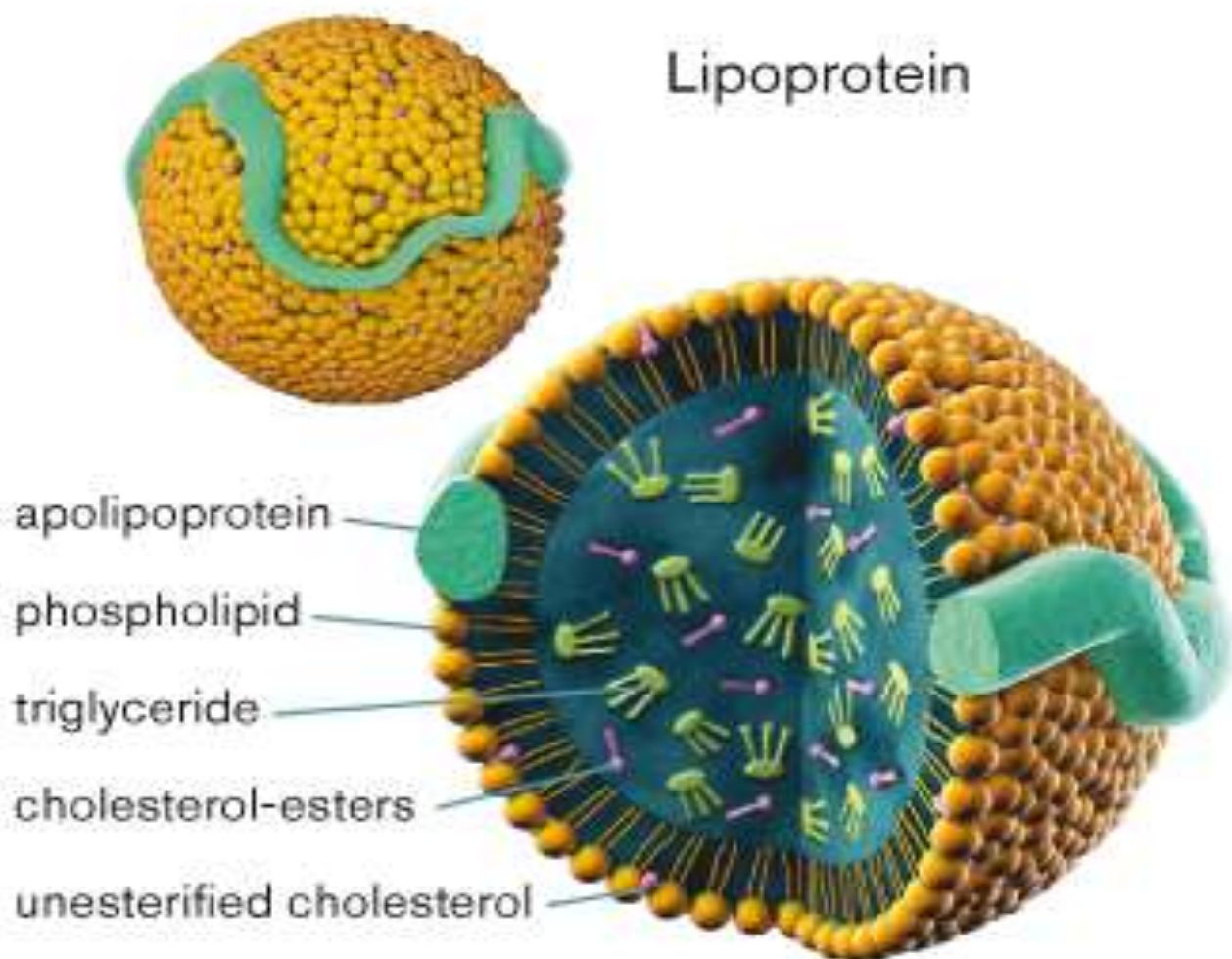
- Pro-brain natriuretic peptide(NT-proBNP)
- This is increased in patients with heart failure
- It has been approved as a marker for acute congestive heart failure
- High serum concentration of brain natriuretic peptide is a marker for congestive cardiac failure

Lipoproteins and their Role in Cardiovascular System

What is lipoprotein?



Lipoprotein



CLASSIFICATION

Chylomicrons

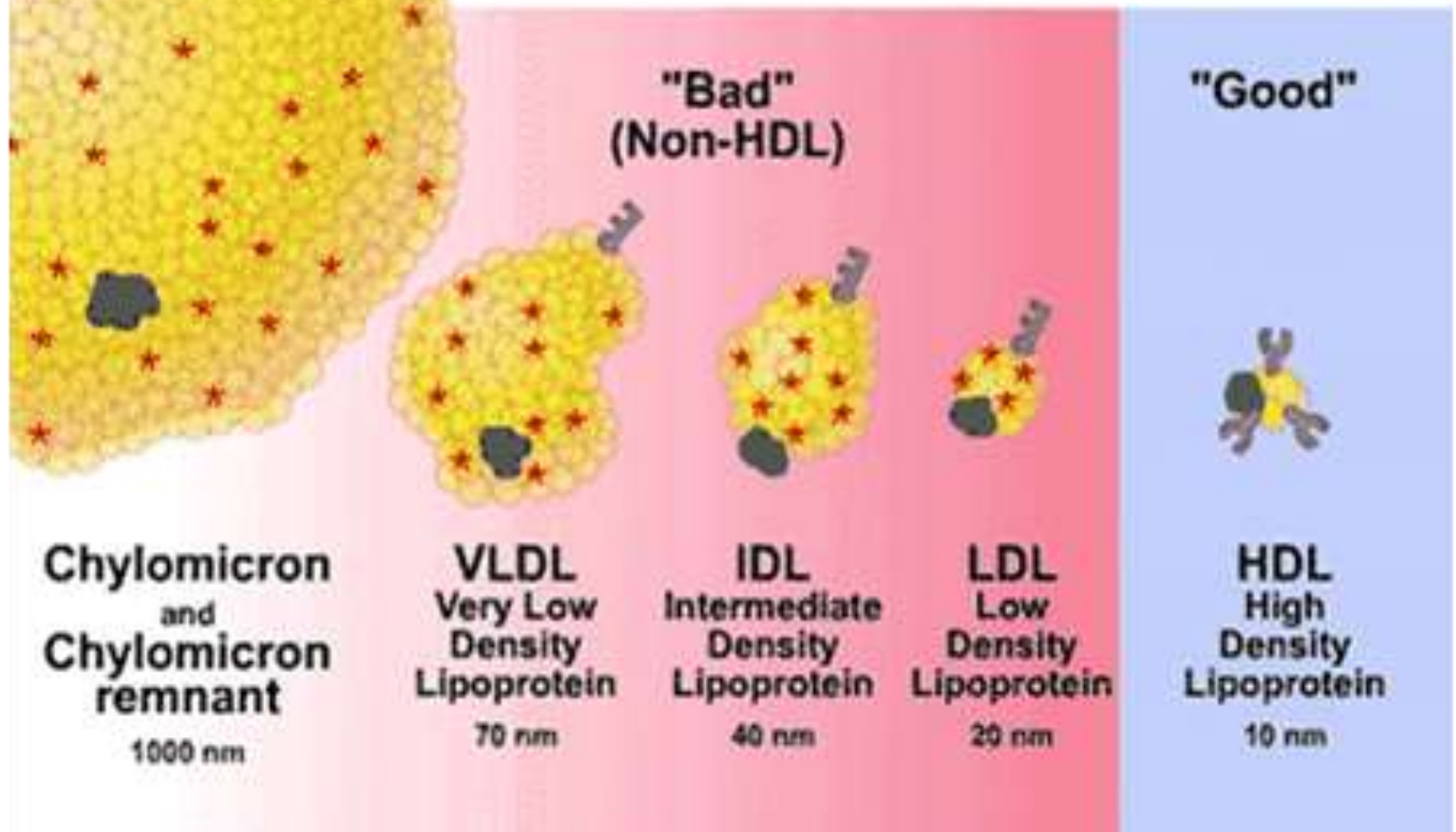
Very low density lipoproteins - VLDL

Intermediate density lipoprotein - IDL

Low density lipoprotein - LDL

High density lipoprotein - HDL

Classification of Lipoproteins



Fat has a density of about 0.88 g/ml, whereas protein has a density of about 1.0 g/ml as consequence, the higher the fat content, the lower the density of the particle

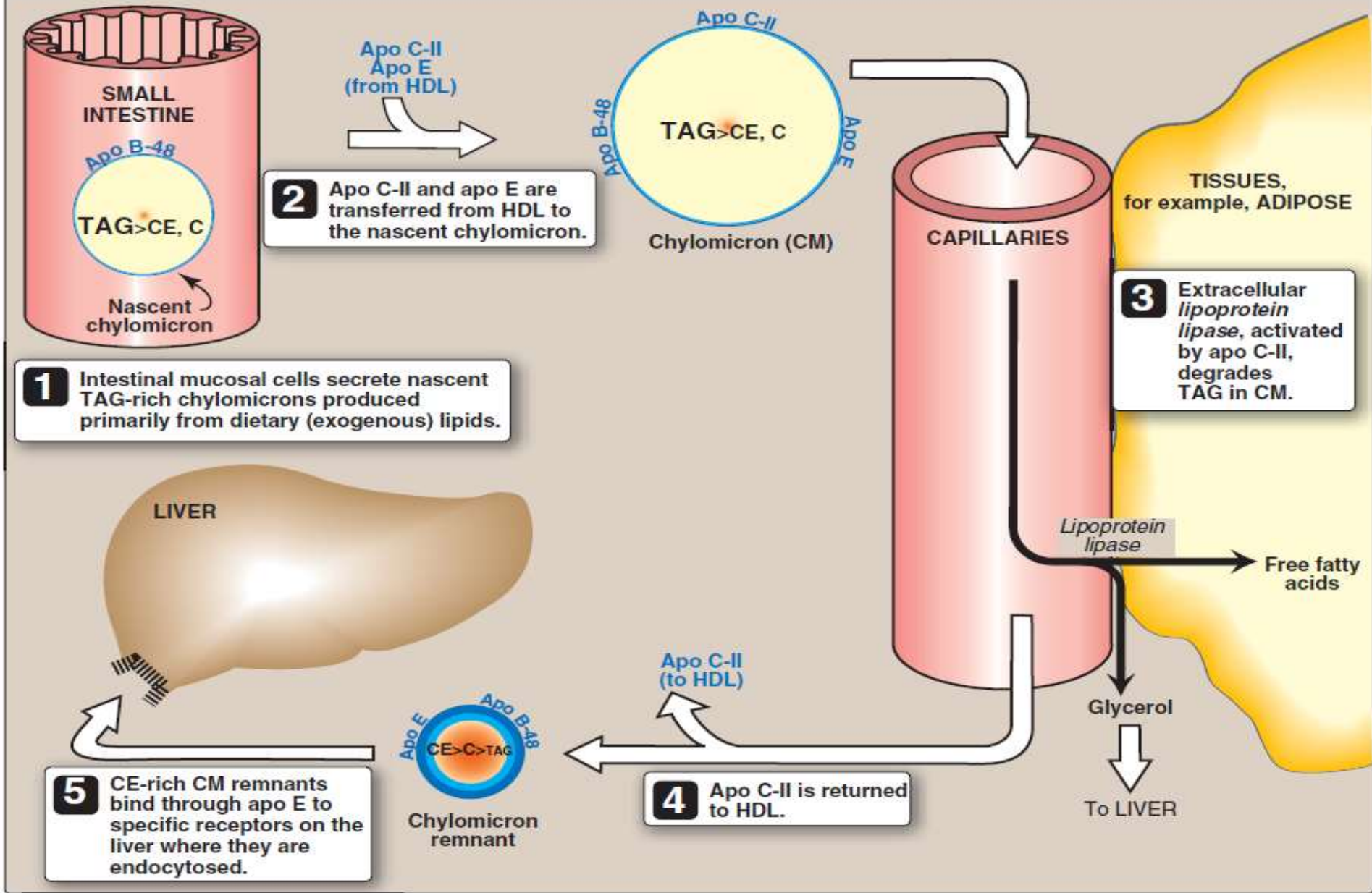
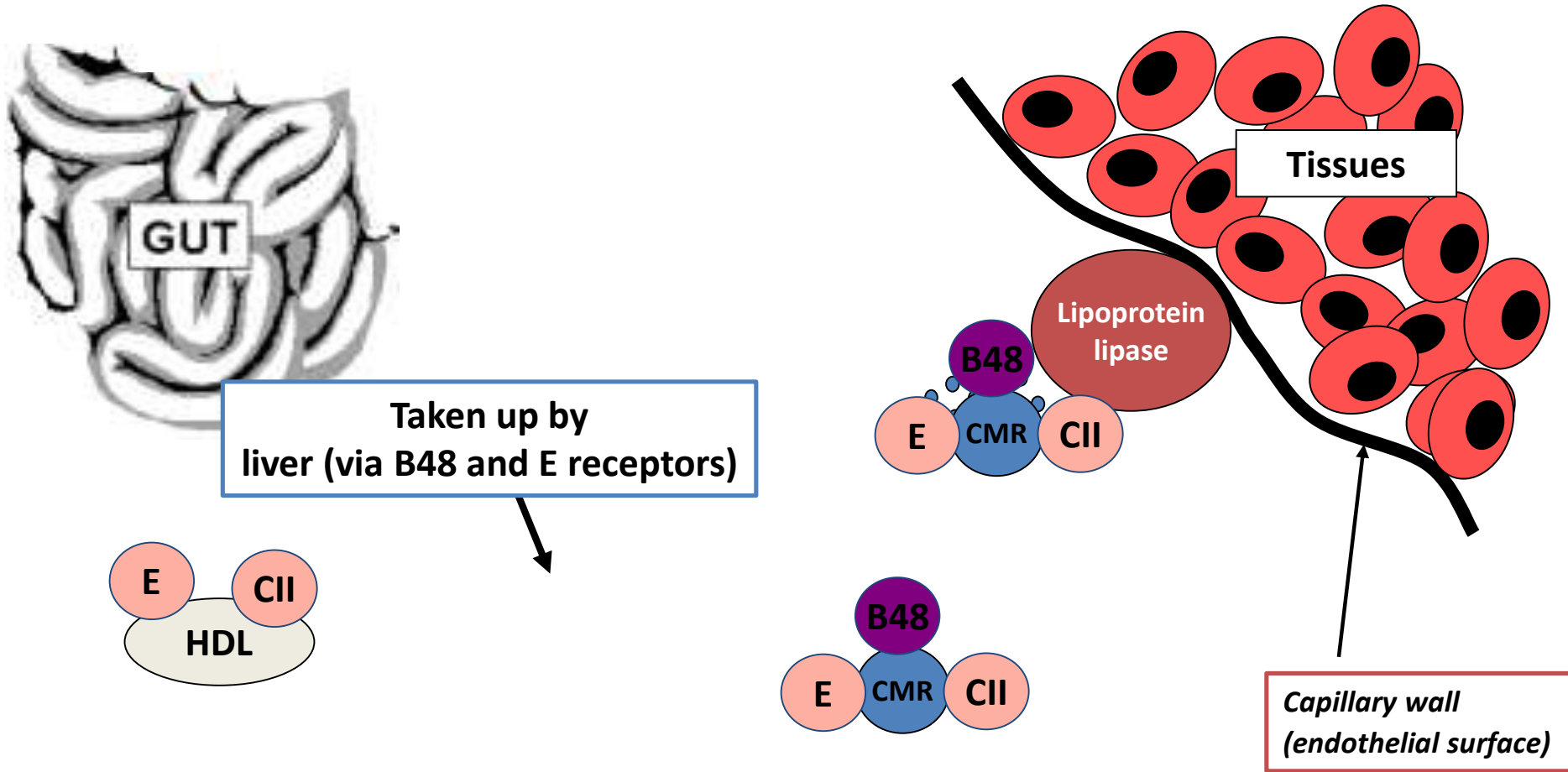


Figure 18.16

Metabolism of chylomicrons. CM = chylomicron; TAG = triacylglycerol; C = cholesterol; CE = cholesteryl esters. Apo B-48, apo C-II, and apo E are apolipoproteins found as specific components of plasma lipoproteins. The lipoproteins are not drawn to scale (see Figure 18.13 for details of the size and density of lipoproteins).

This animation shows how chylomicrons are metabolised once they enter the circulation from the lymphatic system



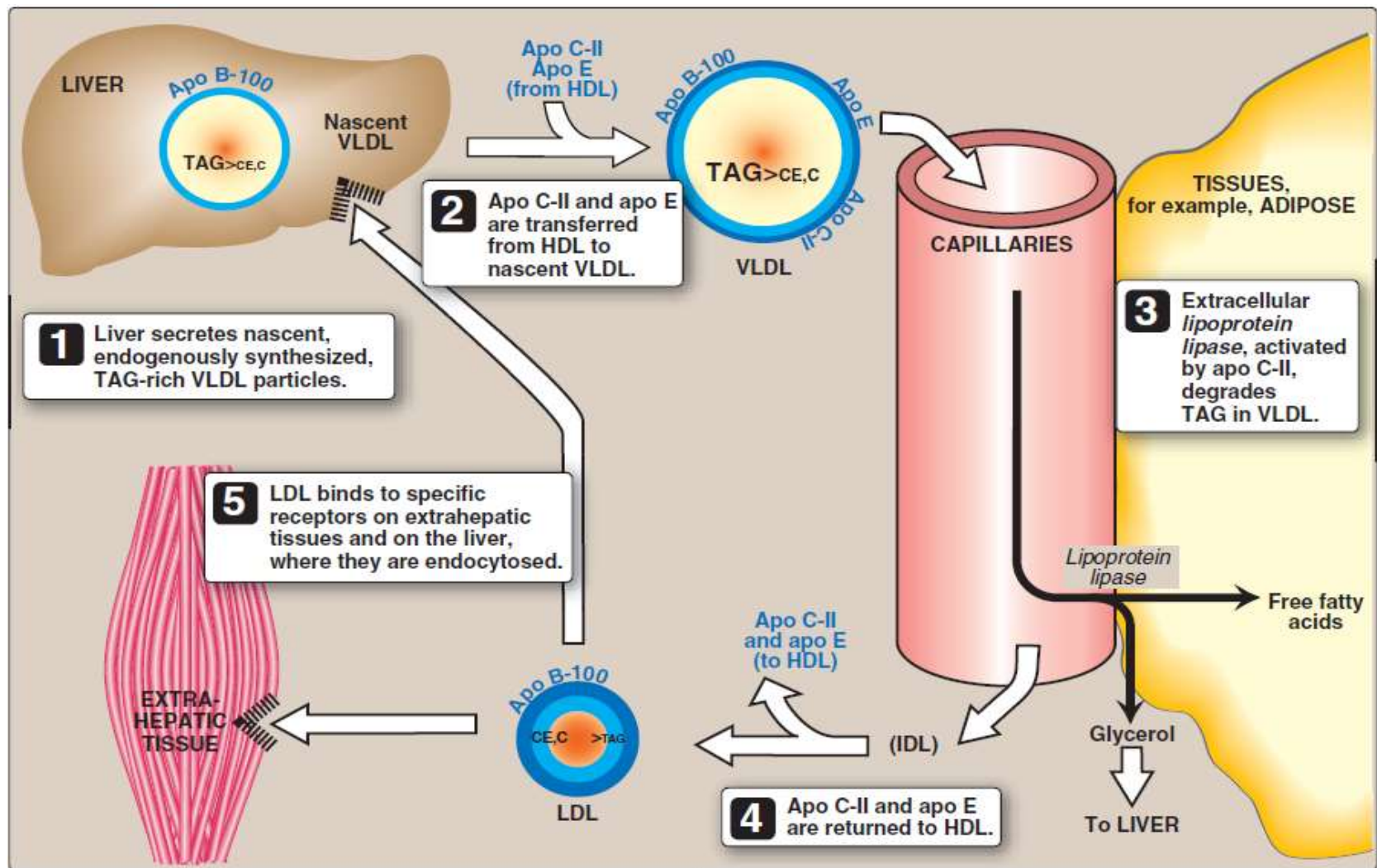
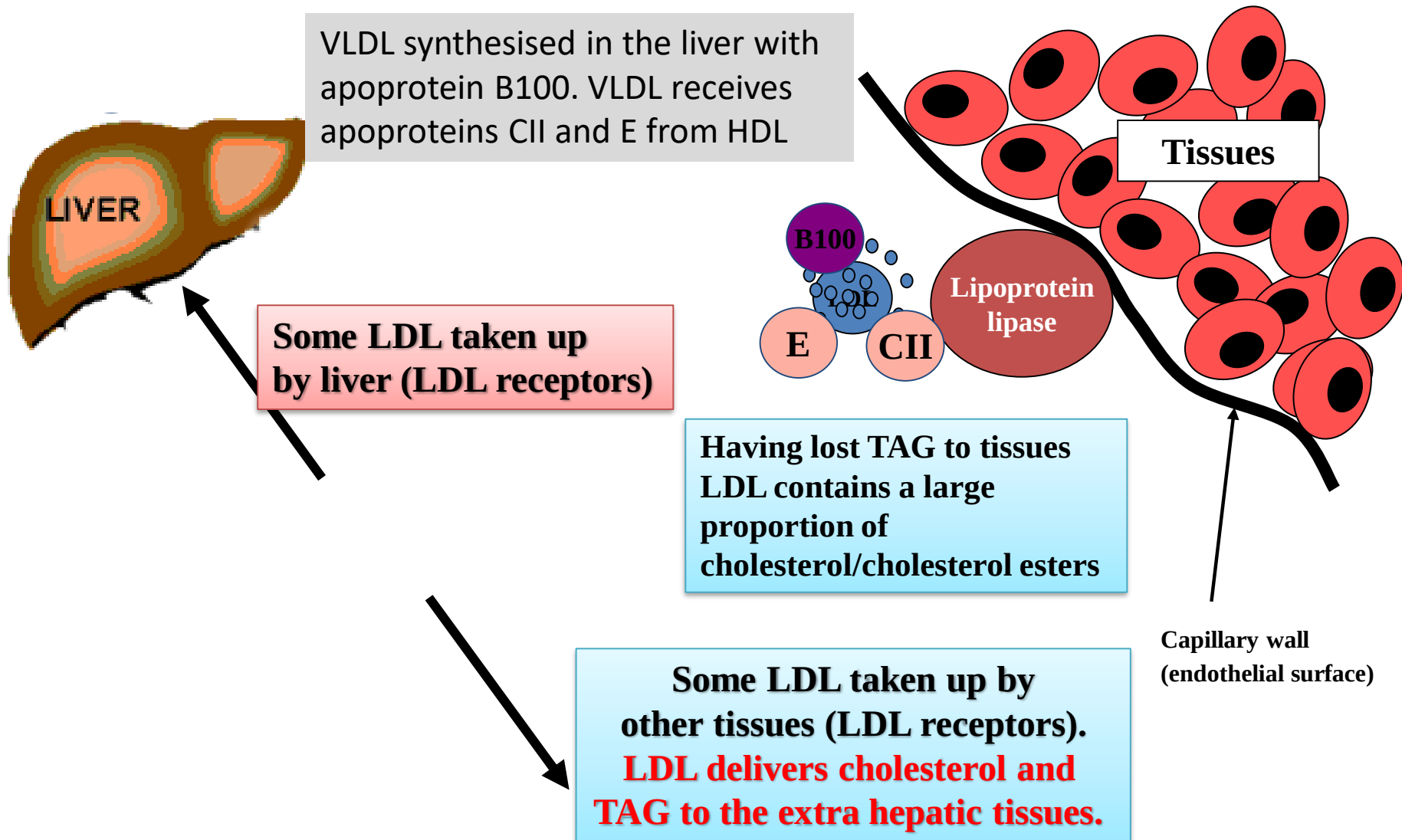


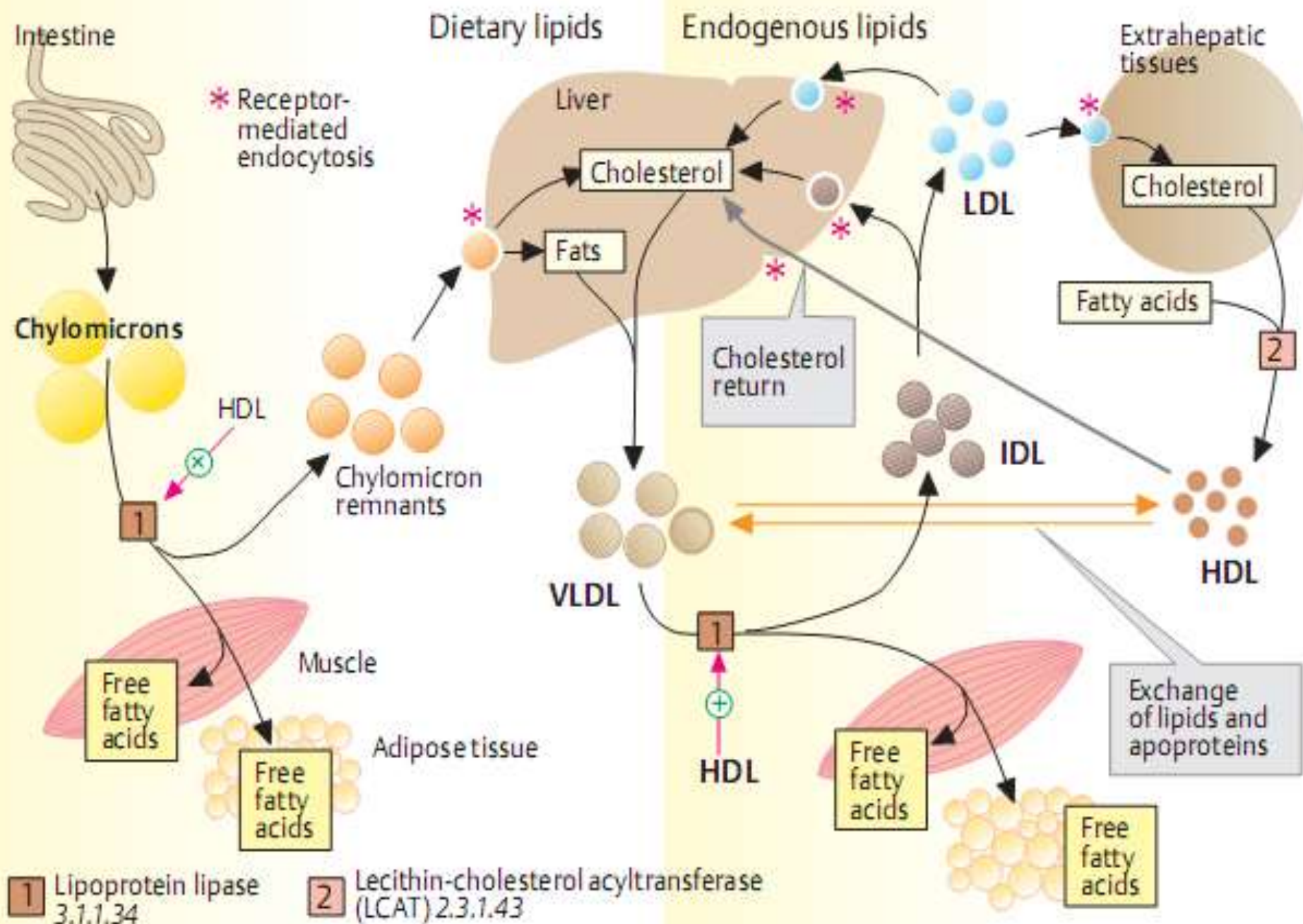
Figure 18.17

Metabolism of VLDL and LDL. TAG = triacylglycerol; VLDL = very-low-density lipoprotein; LDL = low-density-lipoprotein; IDL = intermediate-density lipoprotein; C = cholesterol; CE = cholesteryl esters. Apo B-100, apo C-II, and apo E are apolipoproteins found as specific components of plasma lipoproteins. Lipoproteins are not drawn to scale (see Figure 18.13 for details of the size and density of lipoproteins).

This animation shows how VLDL are metabolised once they enter the circulation from the liver



B. Transport functions



HYPERLIPIDEMIA

- Abnormally elevated level of lipid in blood, such as cholesterol and triglycerides
- these lipids can adhere to the walls of the arteries and restrict blood flow
 - **which creates significant risk of heart attack and stroke**

There are 3 major lipids in our blood

Cholesterol, Triglycerides & Phospholipids

- Cholesterol which is necessary for the synthesis of bile acid vit D, etc...
- Triglycerides : which provides energy to the cell.
- Phospholipids:which is the major component of cell m/brane

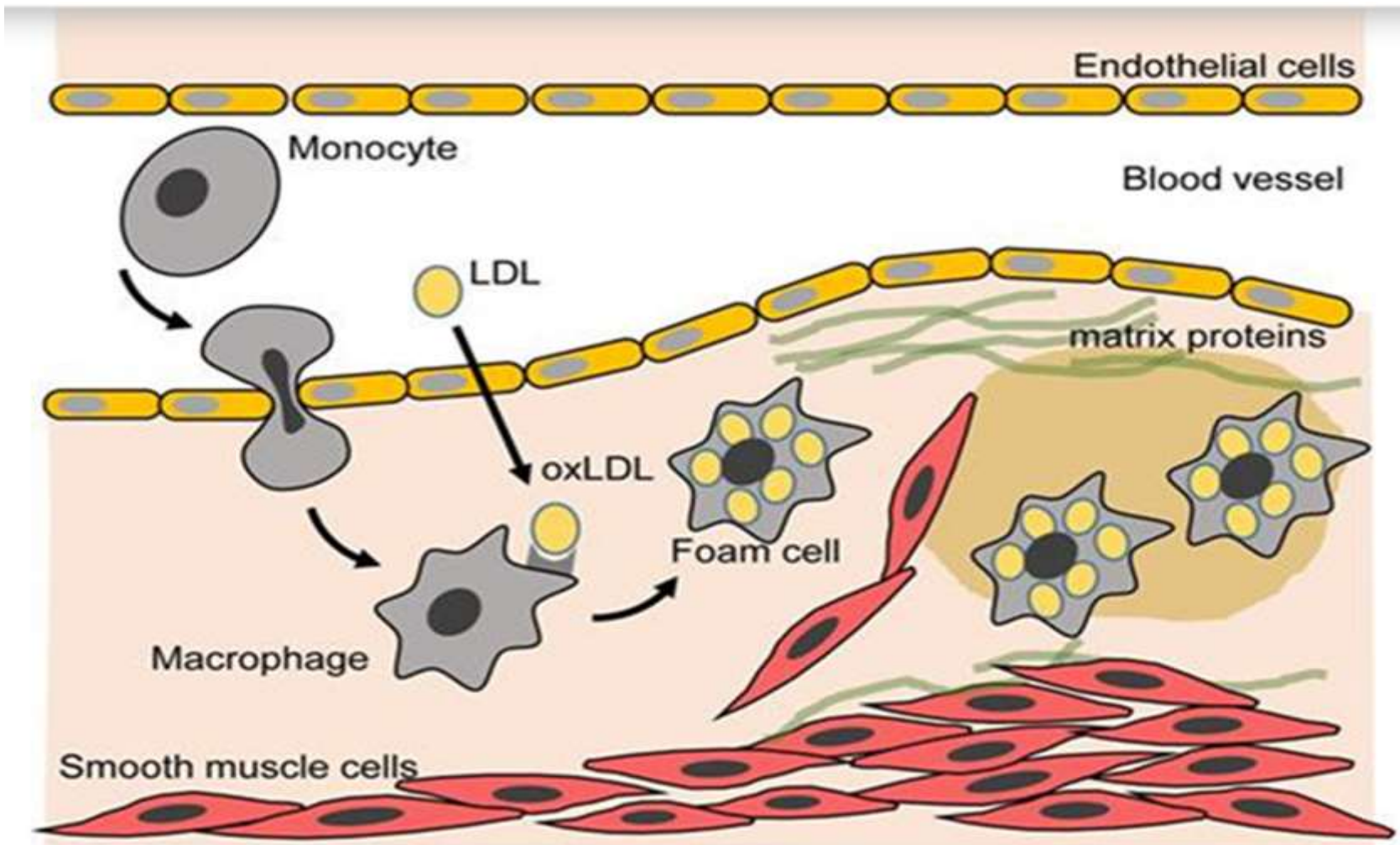
Atherosclerosis...

- Atherosclerosis involves damage to, or dysfunction of, the endothelial cells that form the inner lining of blood vessels, resulting in entry of LDL particles into the vessel wall (Fig).
- Lipids and proteins of the LDL particle undergo oxidative modification by
 - reactive oxygen species (e.g. superoxide, O_2^-), to form oxidized LDL (oxLDL).
- Accumulation of these atherogenic particles attracts macrophages,
- macrophages via 'scavenger' receptors engulf the cholesterol /LDL particles,
 - causing formation of foam cells and subsequent fatty streaks within the endothelium.

Atherosclerosis...

- Various pro-inflammatory mediators are produced during this process which stimulate;
 - smooth muscle cell proliferation, and migration of these cells into the sub endothelial layer
- this process promote inflammation and further the development of atherosclerotic plaques
- The plaque may partially block the lumen of the blood vessel or eventually rupture leading to formation of a thrombus as blood platelets adhere to the exposed sub endothelial collagen

Atherosclerosis...



Lipoproteins and the process of atherosclerosis

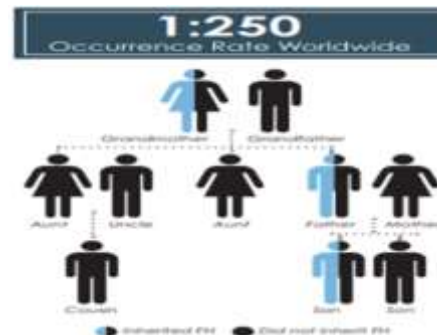
Atherosclerosis...

- Several genetic defects in the LDL receptor and apoprotein genes cause **hyperlipidaemia**
 - and are associated with an increased risk of coronary heart disease, if untreated.

Familial hypercholesterolemia

Familial hypercholesterolemia (FH) is an autosomal dominant genetic condition where,

- the uptake of LDL-C into cells is either decreased or inhibited
- is due to mutations in the LDL receptor
- it is the most common genetic disorder causing
 - premature/early cardiovascular disease (CVD) and death
- It is estimated 1 in 250 has FH, and only about 10% have been identified



Familial hypercholesterolemia ...

- Familial hypercholesterolaemia is a disorder of lipid metabolism that leads to atherosclerosis development
- the hallmark is the increased levels of low-density lipoprotein cholesterol (LDL-C) in blood.
- is an inherited defect,
 - in how the body recycles LDL-C, or “bad” cholesterol.
- can cause narrowing or blocked arteries and plaques, increasing heart attacks and strokes

Familial hypercholesterolemia ...

- Not all types of hypercholesterolemia are equal
- FH exists in individuals even before they are born, and the more serious form can cause narrowing of the blood vessels while the baby is still in the womb.
- The condition will results in lifetime exposure to very high levels of LDL-C

Familial hypercholesterolemia (FH)...

Familial hypercholesterolemia (FH) has two types

- **Heterozygous FH (HeFH):** More common, inherited from one parent.
- This gene is called a dominant gene
- This means that if a person inherits even one of these genes from a parent, they will develop hypercholesterolemia
- In severe cases, LDL cholesterol levels can reach above 190 mg/dL of blood
- Those with one parent with FH have a 50-percent chance of inheriting the condition

Familial hypercholesterolemia (FH)...

- Homozygous FH (HoFH): Rare, inherited from both parents
- People with two gene mutations have extremely high cholesterol levels
- This results in a more severe form of FH, called homozygous FH (occurs in 1 : 250,000)
- If left untreated, a person with homozygous FH may die of heart disease in childhood

Familial hypercholesterolemia (FH)...

- In FH there is a mutation in the gene, that codes for the LDL-receptor, on the cells surface
- The change in the LDL-receptor gene results in abnormal LDL-receptors making
- unable to remove LDL Cholesterol from the blood
- A normal total cholesterol level is approximately less than 200 mg/dL, but
 - people with FH may have a cholesterol level in the 350 to 500 mg/dL range

Familial hypercholesterolemia (FH)...

- Most cases (up to 80%) of familial hypercholesterolemia (FH) are caused by
- mutations in, the LDL receptor gene can reduce
 - the number of LDL receptors produced within cells or decrease the ability of the receptor to bind LDL particles
- Mutations in APO-B disrupt binding of LDL particles to the receptor, while
- Mutations in PCSK9 cause increased degradation of LDL receptor
- These mechanisms lead to elevated LDL-C levels and
 - premature development of atherosclerotic plaque

Familial hypercholesterolemia (FH)...

- Proprotein convertase subtilisin/kexin type 9 (PCSK9) is an enzyme encoded by the PCSK9 gene in humans on chromosome-1.
- It is member of the proprotein convertase family of proteins that activate other proteins
- As with many enzymes, PCSK9 is inactive when first synthesized, because a section of peptide chains blocks their activity; proprotein convertases remove that section to activate the enzyme

Familial hypercholesterolemia (FH)...

- PCSK9 is ubiquitously expressed in many tissues and cell types
- PCSK9 binds to and degrades the receptor for low-density lipoprotein particles (LDL), which typically transport
 - 3,000 to 6,000 fat molecules per particle
- Normally LDL receptor (LDLR), on liver and other cell membranes, binds and initiates ingestion of LDL-particles from extracellular fluid into cells and
 - **targets the complex to lysosomes for destruction**
- Therefore, blocking PCSK9 can lower blood LDL-particle concentrations

Familial hypercholesterolemia (FH)...

- Agents that block PCSK9 can lower LDL particle concentrations.
- PCSK9 promotes degradation of low-density lipoprotein receptor (LDLR) and increasing the risk of cardiovascular disease.

Familial hypercholesterolemia ...

- Mutations in the apoproteins that function as ligands for the LDL receptor (e.g. apo B-100 and apoE) can cause high LDL concentrations and an increased risk of atherosclerosis
- It is worth highlighting that in addition to diet and genetics, there are many other factors that are recognized as risk factors for atherosclerosis
 - age, gender, smoking, high blood pressure, obesity, Type 2 diabetes, stress and physical inactivity

Familial hypercholesterolemia (FH)...

FH diagnosed

FH is usually diagnosed by a blood cholesterol test, called a “lipid panel,” and Genetic testing may be done

Medicines

- A category of medicines, called statins, are the primary drugs to lower LDL-c In addition,
 - ezetimibe and bile acid sequestrants are other medicines used to lower LDL cholesterol
- A new class of medications called PCSK9 Inhibitors are given as injections under the skin

Familial hypercholesterolemia ...

- Sometimes high cholesterol is the result of **unhealthy lifestyle** choices, and that kind is preventable and treatable
- Eating a healthy diet, exercising regularly, not smoking can reducing LDL levels and lowering the risk for heart disease
- Unfortunately for people with FH, they have a hereditary form of hypercholesterolemia, can't be controlled simply through diet and exercise.
- FH is a serious, life-threatening condition and almost always requires medical treatment

Lipoprotein Profile

- Cholesterol levels are measured in milligrams (mg) of cholesterol per deciliter (dL) of blood
 - Desirable or optimal levels for adults are:
 - Total cholesterol: Less than 200 mg/dL.
 - Low Density Lipoprotein (LDL) cholesterol: Less than 100 mg/dL
 - High Density Lipoprotein (HDL): 40 mg/dL or higher.
 - Triglycerides: Less than 150 mg/dL.
- The National Cholesterol Education Program recommends that healthy adults over 20 years of age have their cholesterol levels checked once every 5 years

Lipoprotein Profile...

- LDL-cholesterol can be calculated by the following formula:
- $\text{LDL-cholesterol} = \text{Total cholesterol} - (\text{HDL-cholesterol} + \text{TAG}/5)$
- Cholesterol is present in plasma mainly in LDL (70%; the “bad cholesterol”) and in smaller amounts in HDL (“the good cholesterol”)

Summary of Products Derived from Cholesterol

Cholesterol is not only an essential component of cell membrane, but also a precursor of the following biologically active compounds

1. Vitamin D: Formation of vitamin D₃ occurs by ultraviolet irradiation of the skin. It converts 7-dehydrocholesterol into cholecalciferol.
2. Adrenocortical hormone: The steroid hormones of the adrenal cortex are produced from cholesterol
3. Sex hormones: The male sex hormone, testosterone (19-C), and the female sex hormone, estrogen (18-C) are also derived from cholesterol.
4. Bile acids and bile salts

NB: “Read on Frederickson’s classification of hyperlipoproteinemias”

